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United States Patent	12415985
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Amit; Michal et al.

Culture media, cell cultures and methods of culturing pluripotent stem cells in an undifferentiated state

Abstract

Provided are novel serum-free culture media which comprise basic fibroblast growth factor (bFGF), transforming growth factor beta-3 and ascorbic acid at a concentration of at least about 50 microgram/ml; ascorbic acid at a concentration range of about 400-600 microgram/ml, bFGF at a concentration range of about 50-200 ng/ml, xeno-free serum replacement and a lipid mixture; the IL6RIL6 chimera at a concentration range of about 50-200 picogram per milliliter (pg/ml); or leukemia inhibitory factor (LIF) at a concentration of at least 2000 units/ml; cell cultures comprising same with pluripotent stem cells such as human embryonic stem cells and induced pluripotent stem (iPS) cells, and methods of using same for expanding pluripotent stem cells in an undifferentiated state using two-dimensional or three-dimensional culture systems; and methods of expanding iPS cells in a suspension culture devoid of substrate adherence and cell encapsulation.

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Appl. No.:	17/097023
Filed:	November 13, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210079342 A1	Mar. 18, 2021

Related U.S. Application Data

continuation parent-doc US 13508991 US 9018010 20150428 WO PCT/IL2010/000937 20101111
child-doc US 14696520
division parent-doc US 14696520 20150427 US 10876094 child-doc US 17097023
us-provisional-application US 61272860 20091112

Publication Classification

Int. Cl.: C12N5/071 (20100101); C12N5/0735 (20100101); C12N5/074 (20100101)

U.S. Cl.:

CPC C12N5/0602 (20130101); C12N5/0606 (20130101); C12N5/0696 (20130101);
C12N2500/36 (20130101); C12N2500/90 (20130101); C12N2500/92 (20130101);
C12N2500/98 (20130101); C12N2500/99 (20130101); C12N2501/115 (20130101);
C12N2501/15 (20130101); C12N2501/23 (20130101); C12N2506/45 (20130101)

Field of Classification Search

USPC: None

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Primary Examiner: Pyla; Evelyn Y

Background/Summary

RELATED APPLICATIONS (1) This application is a division of U.S. patent application Ser. No. 14/696,520 filed on Apr. 27, 2015, which is a continuation of U.S. patent application Ser. No. 13/508,991 filed on May 10, 2012, now U.S. Pat. No. 9,018,010, which is a National Phase of PCT Patent Application No. PCT/IL2010/000937 having International Filing Date of Nov. 11, 2010, which claims the benefit of priority of U.S. Provisional Patent Application No. 61/272,860 filed on Nov. 12, 2009. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

SEQUENCE LISTING STATEMENT

(1) The ASCII file, entitled 84991SequenceListing.txt, created on Nov. 11, 2020, comprising 35,204 bytes, submitted concurrently with the filing of this application is incorporated herein by reference.

FIELD AND BACKGROUND OF THE INVENTION

(2) The present invention, in some embodiments thereof, relates to xeno-free culture media which can be used to maintain stem cells in a pluripotent and undifferentiated state, and in some embodiments to defined culture media, cell cultures comprising same and methods using same for culturing pluripotent stem cells in a suspension culture.

(3) The exceptional differentiation potential of human embryonic stem cells (hESCs) underlines them as one of the best models to study early human development, lineage commitment, differentiation processes and to be used for industrial purposes and cell-based therapy.

(4) Induced pluripotent (iPS) cells are somatic cells which are re-programmed to ESC-like cells capable of differentiation into representative tissues of the three embryonic germ layers both in vitro and in vivo. Mouse or human iPS cells were generated by over expression of four transcription factors, c-Myc, Oct4, Klf4 and Sox2 in somatic cells. The iPS cells were shown to form the same colony morphology as ESCs and to express some typical ESCs markers such as Myb, Kit, Gdf3 and Zic3, but less prominently markers such as Dnmt3a, Dnmt3b, Utf1, Tcl1 and the LIF receptor gene, confirming that iPS cells are similar but not identical to ES cells [Takahashi and Yamanaka, 2006; Takahashi et al, 2007; Meissner et al, 2007; Okita et al, 2007]. Yu Junying et al. (Science 318:1917-1920, 2007) found a common gene expression pattern to fibroblasts-derived iPS cells and hESCs.

(5) Further studies revealed that iPS cells could be obtained by transforming somatic cells with Oct4, Sox2, Nanog and Lin28 while omitting the use of the oncogene C-Myc [Yu et al, 2007; Nakagawa et al, 2008]. Improvements of iPS cells derivation methods include the use of plasmids

- instead of viral vectors or derivation without any integration to the genome, which might simplify the future use of iPS cells for clinical applications [Yu J, et al., Science. 2009, 324: 797-801].
- (6) The currently available iPS cells are those derived from embryonic fibroblasts [Takahashi and Yamanaka, 2006; Meissner et al, 2007], fibroblasts formed from hESCs [Park et al, 2008], Fetal fibroblasts [Yu et al, 2007; Park et al, 2008], foreskin fibroblast [Yu et al, 2007; Park et al, 2008], adult dermal and skin tissues [Hanna et al, 2007; Lowry et al, 2008], b-lymphocytes [Hanna et al 2007] and adult liver and stomach cells [Aoi et al, 2008].
- (7) Similarly to hESCs, iPS cells are traditionally cultured with a supportive layer in 2D culture, which allows their continuous growth in the undifferentiated state. For example, iPS cells were cultured on feeder-layers consisting of inactivated mouse embryonic fibroblasts (MEF) or foreskin fibroblasts [Takahashi and Yamanaka 2006, Meissner et al 2007] in the presence of a medium supplemented with fetal bovine serum (FBS). Further improvements of the culturing methods include culturing iPS cells on MEF feeder layers in the presence of a more defined culture medium containing serum replacement and 10 ng/ml of basic fibroblasts growth factor (bFGF) (Park et al., 2008). However, for clinical applications (e.g., cell-based therapy) or industrial purposes, the iPS cells should be cultured in a defined, xeno-free (e.g., animal-free) and a scalable culture system with controlled processes.
- (8) PCT Publication No. WO2007/026353 discloses a well-defined, xeno-free culture media which comprise a TGF-beta isoform or the chimera formed between IL6 and the soluble IL6 receptor (IL6RIL6) for maintaining human embryonic stem cells, in an undifferentiated state in a two-dimensional culture system.
- (9) U.S. Patent Application No. 20050233446 discloses a defined medium which comprises bFGF, insulin and ascorbic acid for maintaining hESCs when cultured on Matrigel™ in an undifferentiated state.
- (10) Ludwig T E., et al., 2006 (Nature Biotechnology, 24: 185-7) discloses the TeSR1 defined medium for culturing hESCs on a matrix composed of Collagen IV, fibronectin, laminin and vitronectin.
- (11) U.S. Patent Application No. 20090029462 discloses methods of expanding pluripotent stem cells in suspension using microcarriers or cell encapsulation.
- (12) PCT Publication No. WO/2008/015682 discloses a method of expanding and maintaining human embryonic stem cells in a suspension culture under culturing conditions devoid of substrate adherence.
- (13) U.S. Patent Application No. 20070155013 discloses a method of growing pluripotent stem cells in suspension using a carrier which adheres to the pluripotent stem cells.
- (14) U.S. Patent Application No. 20080241919 (Parsons et al.) discloses a method of culturing pluripotent stem cells in a suspension culture in a medium which comprises bFGF, insulin and ascorbic acid in a cell culture vessel that includes a cell-free matrix.
- (15) U.S. Patent Application No. 20080159994 (Mantalaris et al.) discloses a method of culturing pluripotent ES cells encapsulated within alginate beads in a three-dimensional culture in a medium which comprises serum replacement and bFGF.
- (16) U.S. Patent Application No. 20070264713 discloses a method of culturing undifferentiated stem cells in suspension on microcarriers in vessels using a conditioned medium.

SUMMARY OF THE INVENTION

- (17) According to an aspect of some embodiments of the present invention there is provided a culture medium being serum-free and xeno-free comprising basic fibroblast growth factor (bFGF), transforming growth factor beta-3 (TGFβ3) and ascorbic acid, wherein a concentration of the ascorbic acid in the culture medium is at least about 50 µg/ml and wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.
- (18) According to an aspect of some embodiments of the present invention there is provided a

culture medium being serum-free and xeno-free comprising ascorbic acid at a concentration range of about 400-600 µg/ml, basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, xeno-free serum replacement and a lipid mixture, wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.

(19) According to an aspect of some embodiments of the present invention there is provided a culture medium being serum-free comprising an IL6RIL6 chimera at a concentration range of about 50-200 picogram per milliliter (pg/ml), wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.

(20) According to an aspect of some embodiments of the present invention there is provided a culture medium being serum-free comprising a leukemia inhibitory factor (LIF) at a concentration of at least 2000 units/ml, wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.

(21) According to an aspect of some embodiments of the present invention there is provided a culture medium comprising basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml and serum replacement, wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in a suspension culture.

(22) According to an aspect of some embodiments of the present invention there is provided a culture medium consisting of a basic medium, ascorbic acid at a concentration range of about 50 µg/ml to about 500 µg/ml, bFGF at a concentration range between about 2 ng/ml to about 20 ng/ml, L-glutamine, and serum replacement.

(23) According to an aspect of some embodiments of the present invention there is provided a culture medium consisting of a basic medium, ascorbic acid at a concentration range of about 50 µg/ml to about 500 µg/ml, bFGF at a concentration range between about 2 ng/ml to about 20 ng/ml, L-glutamine, serum replacement and a lipid mixture.

(24) According to an aspect of some embodiments of the present invention there is provided a cell culture comprising a pluripotent stem cell and the culture medium of the invention.

(25) According to an aspect of some embodiments of the present invention there is provided a method of deriving an embryonic stem cell line, comprising: (a) obtaining an embryonic stem cell from a pre-implantation stage blastocyst, post-implantation stage blastocyst and/or a genital tissue of a fetus; and (b) culturing the embryonic stem cell in the culture medium of the invention; thereby deriving the embryonic stem cell line.

(26) According to an aspect of some embodiments of the present invention there is provided a method of deriving an induced pluripotent stem cell line, comprising: (a) inducing a somatic cell to a pluripotent stem cell; and (b) culturing the pluripotent stem cell in the culture medium of the invention; thereby deriving the induced pluripotent stem cell line.

(27) According to an aspect of some embodiments of the present invention there is provided a method of expanding and maintaining pluripotent stem cells in an undifferentiated state, the method comprising culturing the pluripotent stem cells in the culture medium of the invention, thereby expanding and maintaining the pluripotent stem cells in the undifferentiated state.

(28) According to an aspect of some embodiments of the present invention there is provided a method of expanding and maintaining pluripotent stem cells in an undifferentiated state, the method comprising culturing the pluripotent stem cells in a culture medium being serum-free, feeder-free, matrix-free and protein carrier-free and comprising basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state.

(29) According to an aspect of some embodiments of the present invention there is provided a method of expanding pluripotent stem cells and maintaining the pluripotent stem cells in an undifferentiated state, the method comprising culturing the pluripotent stem cells on a feeder cell layer in a serum-free and xeno-free culture medium, the culture medium comprises basic fibroblast

growth factor (bFGF) transforming growth factor beta-3 (TGF β 3) and ascorbic acid, wherein a concentration of the ascorbic acid in the culture medium is at least 50 μ g/ml and wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state, thereby expanding and maintaining the stem cells in the undifferentiated state.

(30) According to an aspect of some embodiments of the present invention there is provided a method of expanding pluripotent stem cells and maintaining the pluripotent stem cells in an undifferentiated state, the method comprising culturing the pluripotent stem cells on a feeder cell layer in a serum-free and xeno-free culture medium, the culture medium comprises ascorbic acid at a concentration range of about 400-600 μ g/ml, basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, xeno-free serum replacement and a lipid mixture, wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state, thereby expanding and maintaining the stem cells in the undifferentiated state.

(31) According to an aspect of some embodiments of the present invention there is provided a method of expanding induced pluripotent stem (iPS) cells and maintaining the iPS cells in an undifferentiated state, the method comprising culturing the iPS cells in a suspension culture under culturing conditions devoid of substrate adherence and devoid of cell encapsulation and which allow expansion of the iPS cells in the undifferentiated state, thereby expanding and maintaining the iPS cells in the undifferentiated state.

(32) According to an aspect of some embodiments of the present invention there is provided a method of generating lineage-specific cells from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells according to the method of the invention, to thereby obtain expanded, undifferentiated stem cells; (b) subjecting the expanded, undifferentiated stem cells to culturing conditions suitable for differentiating and/or expanding lineage specific cells; thereby generating the lineage-specific cells from the pluripotent stem cells.

(33) According to an aspect of some embodiments of the present invention there is provided a method of generating embryoid bodies from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells according to the method of the invention, to thereby obtain expanded, undifferentiated pluripotent stem cells; and (b) subjecting the expanded, undifferentiated pluripotent stem cells to culturing conditions suitable for differentiating the stem cells to embryoid bodies; thereby generating the embryoid bodies from the pluripotent stem cells.

(34) According to an aspect of some embodiments of the present invention there is provided a method of generating lineage-specific cells from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells according to the method of the invention, to thereby obtain expanded, undifferentiated pluripotent stem cells; (b) subjecting the expanded, undifferentiated pluripotent stem cells to culturing conditions suitable for differentiating the expanded, undifferentiated stem cells to embryoid bodies; and (c) subjecting cells of the embryoid bodies to culturing conditions suitable for differentiating and/or expanding lineage specific cells; thereby generating the lineage-specific cells from the pluripotent stem cells.

(35) According to some embodiments of the invention, the cell culture is feeder cells free.

(36) According to some embodiments of the invention, the culture medium is capable of expanding the pluripotent stem cells in an undifferentiated state when cultured in a suspension culture.

(37) According to some embodiments of the invention, the stem cells are embryonic stem cells.

(38) According to some embodiments of the invention, the stem cells are induced pluripotent stem (iPS) cells.

(39) According to some embodiments of the invention, the embryonic stem cells are human embryonic stem cells.

(40) According to some embodiments of the invention, the induced pluripotent stem cells are human induced pluripotent stem cells.

(41) According to some embodiments of the invention, the culture medium is capable of expanding the pluripotent stem cells in an undifferentiated state.

- (42) According to some embodiments of the invention, the culture medium further comprises basic fibroblast growth factor (bFGF).
- (43) According to some embodiments of the invention, the culture medium further comprises serum replacement.
- (44) According to some embodiments of the invention, a concentration of the TGF β 3 in the culture medium is at least about 0.5 ng/ml.
- (45) According to some embodiments of the invention, a concentration of the TGF β 3 in the culture medium is about 2 ng/ml.
- (46) According to some embodiments of the invention, a concentration of the bFGF in the culture medium is at least about 5 ng/ml.
- (47) According to some embodiments of the invention, a concentration of the bFGF in the culture medium is in the range of about 5 ng/ml to about 200 ng/ml.
- (48) According to some embodiments of the invention, a concentration of the ascorbic acid in the culture medium is in the range of about 400 microgram/milliliter (μ g/ml) to about 600 μ g/ml.
- (49) According to some embodiments of the invention, a concentration of the ascorbic acid in the culture medium is about 500 μ g/m (microgram/milliliter).
- (50) According to some embodiments of the invention, the culturing is effected on a matrix.
- (51) According to some embodiments of the invention, the matrix comprises an extracellular matrix.
- (52) According to some embodiments of the invention, the extracellular matrix is selected from the group consisting of a fibronectin matrix, a laminin matrix, and a foreskin fibroblast matrix.
- (53) According to some embodiments of the invention, the matrix is xeno-free.
- (54) According to some embodiments of the invention, the feeder cell layer is xeno-free.
- (55) According to some embodiments of the invention, the feeder cell layer comprises foreskin fibroblast cells.
- (56) According to some embodiments of the invention, the bFGF is at a concentration range of about 0.1 ng/ml to about 500 ng/ml, the TGF β 3 is at a concentration range of about 0.1 ng/ml to about 20 ng/ml, the ascorbic acid is at a concentration range of about 50 μ g/ml to about 5000 μ g/ml.
- (57) According to some embodiments of the invention, the bFGF is at a concentration range of about 5 ng/ml to about 150 ng/ml, the TGF β 3 is at a concentration range of about 0.5 ng/ml to about 5 ng/ml, the ascorbic acid is at a concentration range of about 400 μ g/ml to about 600 μ g/ml.
- (58) According to some embodiments of the invention, the culture medium further comprising serum replacement.
- (59) According to some embodiments of the invention, the serum replacement is xeno free.
- (60) According to some embodiments of the invention, the culture medium further comprising a lipid mixture.
- (61) According to some embodiments of the invention, the culture medium further comprising sodium bicarbonate at a concentration of about 5% to about 10%.
- (62) According to some embodiments of the invention, the lipid mixture is at a concentration of about 1%.
- (63) According to some embodiments of the invention, the concentration of the IL6RIL6 chimera is about 100 pg/ml.
- (64) According to some embodiments of the invention, the concentration of the LIF is in a range of about 2000-4000 units/ml.
- (65) According to some embodiments of the invention, the culturing is effected in a suspension culture.
- (66) According to some embodiments of the invention, the culture medium is devoid of TGF β 3.
- (67) According to some embodiments of the invention, the culture medium comprises no more than 0.1 ng/ml of TGF β 3.

- (68) According to some embodiments of the invention, a culture medium of the suspension culture is serum-free and feeder cell-free.
- (69) According to some embodiments of the invention, the culture medium being serum-free and devoid of animal contaminants.
- (70) According to some embodiments of the invention, the concentration of said bFGF is about 100 ng/ml.
- (71) According to some embodiments of the invention, the culture medium comprises an IL6RIL6 chimera at a concentration range of about 50-200 picograms per milliliter (pg/ml), wherein the culture medium is capable of maintaining the iPS cells in an undifferentiated state in the absence of feeder cell support.
- (72) According to some embodiments of the invention, the culture medium comprises leukemia inhibitory factor (LIF) at a concentration of at least 2000 units/ml, wherein the culture medium is capable of maintaining the iPS cells in an undifferentiated state in the absence of feeder cell support.
- (73) According to some embodiments of the invention, the culture medium comprises basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml.
- (74) According to some embodiments of the invention, the culture medium comprises an IL6RIL6 chimera at a concentration range of about 50-200 nanogram per milliliter (ng/ml).
- (75) According to some embodiments of the invention, the culture medium further comprises basic fibroblast growth factor (bFGF).
- (76) According to some embodiments of the invention, the culture medium is protein carrier-free.
- (77) According to some embodiments of the invention, expanding comprises obtaining at least about 8×10^6 cells from a single pluripotent stem cell following about 1 month.
- (78) According to some embodiments of the invention, the pluripotent stem cells cultured in the culture medium exhibits a normal chromosomal karyotype following at least 2 passages.
- (79) According to some embodiments of the invention, the pluripotent stem cells exhibits a doubling time of at least 20 hours.
- (80) According to some embodiments of the invention, maintaining is for at least 5 passages.
- (81) Unless otherwise defined, all technical and/or scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention pertains. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of embodiments of the invention, exemplary methods and/or materials are described below. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and are not intended to be necessarily limiting.
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Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

- (1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.
- (2) Some embodiments of the invention are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of embodiments of the invention. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the invention may be practiced.
- (3) In the drawings:
- (4) FIGS. 1A-1C are photographs depicting colony morphology of iPS cells cultured on a xeno-free

two-dimensional culture system in the presence of the novel xeno-free (e.g., animal-free, devoid of animal contamination) culture media according to some embodiments of the invention. J1.2-3 were cultured with human foreskin fibroblast (HFF) supportive layers while using the following animal-serum free culture medium: FIG. 1A—medium HA70 for 6 passages; FIG. 1B—medium HA40/4 for 6 passages; and FIG. 1C—medium D2 for 16 passages.

(5) FIGS. 2A-2C are photographs depicting immuno-fluorescence staining of iPS cells with markers of pluripotency. J1.2-3 and iF4 iPS cells were cultured on a xeno-free two-dimensional culture system (HFF) in the presence of the animal-serum free culture medium HA77 for at least 10 passages and were then stained with the following markers of undifferentiated markers: FIG. 2A—J1.2-3 iPS cells stained with Oct4; FIG. 2B—iF4 iPS cells stained with SSEA4; and FIG. 2C—iF4 iPS cells stained with TRA-1-81.

(6) FIGS. 3A-3C are photographs depicting the morphology of the J1.2-3 iPS cell line from HFF when cultured in suspension in the following xeno-free culture media for the indicated passages. FIG. 3A—J1.2-3 iPS cells cultured in the yFL3 medium for 16 passages; FIG. 3B—J1.2-3 iPS cells cultured in the CM100F medium for 13 passages; FIG. 3C—J1.2-3 iPS cells cultured in the yF100 medium for 8 passages. Note that while cultured in suspension the iPS cells create sphere like structure containing undifferentiated cells.

(7) FIG. 4 is a photograph depicting the morphology of the J1.2-3 iPS cells when cultured on mouse embryonic fibroblasts (MEF) after an extended culturing period in a suspension culture. J1.2-3 cells were cultured for 37 passages in suspension in the CM100F medium, following which they were re-cultured with MEFs, and form typical iPS colony morphology 24 hours post their culture with MEFs.

(8) FIGS. 5A-5C are photographs depicting immuno-fluorescence staining of iPS cells with markers of pluripotency. J1.2-3 cells were cultured in suspension using medium CM100F for more than 20 passages and were then stained with markers of undifferentiated stem cells. FIG. 5A—TRA-1-81; FIG. 5B—TRA-1-60; FIG. 5C—SSEA4.

(9) FIGS. 6A-6D are photographs depicting immunostaining of iPS cells with markers of pluripotency. J1.2-3 cells were cultured in suspension using the CM100F medium for at least 30 passages and then were transferred into spinner flasks and were cultured for additional 30 days, following which the cells were stained with markers of undifferentiated stem cells. FIG. 6A—Oct4; FIG. 6B—TRA-1-81; FIG. 6C—TRA-1-60; and FIG. 6D—SSEA4.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(10) The present invention, in some embodiments thereof, relates to novel culture media, cell cultures comprising same and methods utilizing same for maintaining pluripotent stem cells in a proliferative, pluripotent and undifferentiated state and, more particularly, but not exclusively, to methods of expanding hESCs and induced pluripotent stem (iPS) cells in suspension cultures or two-dimensional culture systems while maintaining the cells in a proliferative, pluripotent and undifferentiated state.

(11) Before explaining at least one embodiment of the invention in detail, it is to be understood that the invention is not necessarily limited in its application to the details set forth in the following description or exemplified by the Examples. The invention is capable of other embodiments or of being practiced or carried out in various ways.

(12) The present inventors have designed following laborious experimentations defined culture media, which are serum-free and xeno-free (e.g., devoid of animal contaminants) and which can maintain pluripotent stem cells such as human iPS and ESCs in an undifferentiated state in the absence of feeder cell support while preserving their pluripotent potential to differentiate into all three embryonic germ layers.

(13) Thus, as shown in the Examples section which follows, hESCs and iPS cells were cultured in an undifferentiated state on two-dimensional culture systems which are either feeder-layer free (e.g., on a synthetic matrix; Example 1) or xeno-free feeder layer-based (e.g., foreskin fibroblasts;

FIGS. 1A-C and 2A-C, Example 2) in the presence of serum-free, xeno-free and defined culture media (e.g., mHA40/4, HA75, HA76, HA77, HA78 or HA74). While in culture, the pluripotent stem cells exhibit undifferentiated morphology, as well as morphological and molecular characteristics typical to iPS or hESCs such as normal karyotype, expression of markers of pluripotency (e.g., Oct4, SSEA4, TRA-1-81, TRA-1-60), and ability to differentiate into all three embryonic germ layers both in vitro (by formation of embryoid bodies after at least 28 passages) and in vivo (by formation of teratomas after at least 31 passages).

(14) As used herein the phrase “pluripotent stem cells” refers to cells which are capable of differentiating into cells of all three embryonic germ layers (i.e., endoderm, ectoderm and mesoderm). According to some embodiments of the invention, the phrase “pluripotent stem cells” encompasses embryonic stem cells (ESCs) and induced pluripotent stem cells (iPS cells).

(15) The phrase “embryonic stem cells” may comprise cells which are obtained from the embryonic tissue formed after gestation (e.g., blastocyst) before implantation (i.e., a pre-implantation blastocyst), extended blastocyst cells (EBCs) which are obtained from a post-implantation/pre-gastrulation stage blastocyst (see WO2006/040763] and embryonic germ (EG) cells which are obtained from the genital tissue of a fetus any time during gestation, preferably before 10 weeks of gestation.

(16) According to some embodiments of the invention, the pluripotent stem cells of the invention are embryonic stem cells, such as from a human or primate (e.g., monkey) origin.

(17) The embryonic stem cells of the invention can be obtained using well-known cell-culture methods. For example, human embryonic stem cells can be isolated from human blastocysts. Human blastocysts are typically obtained from human in vivo preimplantation embryos or from in vitro fertilized (IVF) embryos. Alternatively, a single cell human embryo can be expanded to the blastocyst stage. For the isolation of human ES cells the zona pellucida is removed from the blastocyst and the inner cell mass (ICM) is isolated by immunosurgery, in which the trophectoderm cells are lysed and removed from the intact ICM by gentle pipetting. The ICM is then plated in a tissue culture flask containing the appropriate medium which enables its outgrowth. Following 9 to 15 days, the ICM derived outgrowth is dissociated into clumps either by a mechanical dissociation or by an enzymatic degradation and the cells are then re-plated on a fresh tissue culture medium. Colonies demonstrating undifferentiated morphology are individually selected by micropipette, mechanically dissociated into clumps, and re-plated. Resulting ES cells are then routinely split every 4-7 days. For further details on methods of preparation human ES cells see Thomson et al., [U.S. Pat. No. 5,843,780; Science 282: 1145, 1998; Curr. Top. Dev. Biol. 38: 133, 1998; Proc. Natl. Acad. Sci. USA 92: 7844, 1995]; Bongso et al., [Hum Reprod 4: 706, 1989]; and Gardner et al., [Fertil. Steril. 69: 84, 1998].

(18) It will be appreciated that commercially available stem cells can also be used with this aspect of the present invention. Human ES cells can be purchased from the NIH human embryonic stem cells registry ([www\[.\]dotescr\[.\]dotnih\[.\]dotgov](http://www.nih.gov)). Non-limiting examples of commercially available embryonic stem cell lines are BG01, BG02, BG03, BG04, CY12, CY30, CY92, CY10, TE03, TE04 and TE06.

(19) Extended blastocyst cells (EBCs) can be obtained from a blastocyst of at least nine days post fertilization at a stage prior to gastrulation. Prior to culturing the blastocyst, the zona pellucida is digested [for example by Tyrode's acidic solution (Sigma Aldrich, St Louis. MO, USA)] so as to expose the inner cell mass. The blastocysts are then cultured as whole embryos for at least nine and no more than fourteen days post fertilization (i.e., prior to the gastrulation event) in vitro using standard embryonic stem cell culturing methods.

(20) Embryonic germ (EG) cells are prepared from the primordial germ cells obtained from fetuses of about 8-11 weeks of gestation (in the case of a human fetus) using laboratory techniques known to anyone skilled in the arts. The genital ridges are dissociated and cut into small chunks which are thereafter disaggregated into cells by mechanical dissociation. The EG cells are then grown in

tissue culture flasks with the appropriate medium. The cells are cultured with daily replacement of medium until a cell morphology consistent with EG cells is observed, typically after 7-30 days or 1-4 passages. For additional details on methods of preparation human EG cells see Shamblott et al., [Proc. Natl. Acad. Sci. USA 95: 13726, 1998] and U.S. Pat. No. 6,090,622.

(21) The phrase “induced pluripotent stem (iPS) cell” (or embryonic-like stem cell) as used herein refers to a proliferative and pluripotent stem cell which is obtained by de-differentiation of a somatic cell (e.g., an adult somatic cell).

(22) According to some embodiments of the invention, the iPS cell is characterized by a proliferative capacity which is similar to that of ESCs and thus can be maintained and expanded in culture for an almost unlimited time.

(23) IPS cells can be endowed with pluripotency by genetic manipulation which re-program the cell to acquire embryonic stem cells characteristics. For example, the iPS cells of the invention can be generated from somatic cells by induction of expression of Oct-4, Sox2, Klf4 and c-Myc in a somatic cell essentially as described in Takahashi and Yamanaka, 2006, Takahashi et al, 2007, Meissner et al, 2007, and Okita et al, 2007). Additionally or alternatively, the iPS cells of the invention can be generated from somatic cells by induction of expression of Oct4, Sox2, Nanog and Lin28 essentially as described in Yu et al, 2007, and Nakagawa et al, 2008. It should be noted that the genetic manipulation (re-programming) of the somatic cells can be performed using any known method such as using plasmids or viral vectors, or by derivation without any integration to the genome [Yu J. et al., Science. 2009, 324: 797-801].

(24) The iPS cells of the invention can be obtained by inducing de-differentiation of embryonic fibroblasts [Takahashi and Yamanaka, 2006; Meissner et al, 2007], fibroblasts formed from hESCs [Park et al, 2008], Fetal fibroblasts [Yu et al, 2007; Park et al, 2008], foreskin fibroblast [Yu et al, 2007; Park et al, 2008], adult dermal and skin tissues [Hanna et al, 2007; Lowry et al, 2008], b-lymphocytes [Hanna et al 2007] and adult liver and stomach cells [Aoi et al, 2008].

(25) IPS cell lines are also available via cell banks such as the WiCell bank. Non-limiting examples of commercially available iPS cell lines include the iPS foreskin clone 1 [WiCell Catalogue No. iPS(foreskin)-1-DL-1], the iPSIMR90 clone 1 [WiCell Catalogue No. iPS(IMR90)-1-DL-1], and the iPSIMR90 clone 4 [WiCell Catalogue No. iPS(IMR90)-4-DL-1].

(26) According to some embodiments of the invention, the induced pluripotent stem cells are human induced pluripotent stem cells.

(27) As used herein the phrase “culture medium” refers to a liquid substance used to support the growth of pluripotent stem cells and maintain them in an undifferentiated state. The culture medium used by the invention according to some embodiments can be a water-based medium which includes a combination of substances such as salts, nutrients, minerals, vitamins, amino acids, nucleic acids, proteins such as cytokines, growth factors and hormones, all of which are needed for cell proliferation and are capable of maintaining the pluripotent stem cells in an undifferentiated state. For example, a culture medium according to an aspect of some embodiments of the invention can be a synthetic tissue culture medium such as the Ko-DMEM (Gibco-Invitrogen Corporation products, Grand Island, NY, USA), DMEM/F12 (Biological Industries, Biet HaEmek, Israel), Mab ADCB medium (HyClone, Utah, USA) supplemented with the necessary additives as is further described hereinunder.

(28) The phrase “feeder cell support” as used herein refers to the ability of a feeder cell (e.g., fibroblasts) to maintain pluripotent stem cells in a proliferative and undifferentiated state when the pluripotent stem cells are co-cultured on the feeder cells or when the pluripotent stem cells are cultured on a matrix (e.g., an extracellular matrix, a synthetic matrix) in the presence of a conditioned medium generated by the feeder cells. The support of the feeder cells depends on the structure of the feeder cells while in culture (e.g., the three dimensional matrix formed by culturing the feeder cells in a tissue culture plate), function of the feeder cells (e.g., the secretion of growth factors, nutrients and hormones by the feeder cells, the growth rate of the feeder cells, the

expansion ability of the feeder cells before senescence) and/or the attachment of the pluripotent stem cells to the feeder cell layer(s).

(29) The phrase “absence of feeder cell support” as used herein refers to a culture medium and/or a cell culture being devoid of feeder cells and/or a conditioned medium generated thereby.

(30) As used herein the phrase “serum-free” refers to being devoid of a human or an animal serum.

(31) It should be noted that the function of serum in culturing protocols is to provide the cultured cells with an environment similar to that present in vivo (i.e., within the organism from which the cells are derived, e.g., a blastocyst of an embryo). However, the use of serum, which is derived from either an animal source (e.g., bovine serum) or a human source (human serum), is limited by the significant variations in serum components between the donor individuals (from which the serum is obtained) and the risk of having xeno contaminants (in case of an animal serum is used).

(32) According to some embodiments of the invention, the serum-free culture medium does not comprise serum or portions thereof.

(33) According to some embodiments of the invention, the serum-free culture medium of the invention is devoid of serum albumin (e.g., albumin which is purified from human serum or animal serum).

(34) According to some embodiments of the invention the culture medium comprises serum replacement.

(35) As used herein the phrase “serum replacement” refers to a defined formulation, which substitutes the function of serum by providing pluripotent stem cells with components needed for growth and viability.

(36) Various serum replacement formulations are known in the art and are commercially available.

(37) For example, GIBCO™ KNOCKOUT™ Serum Replacement (Gibco-Invitrogen Corporation, Grand Island, NY USA, Catalogue No. 10828028) is a defined serum-free formulation optimized to grow and maintain undifferentiated ES cells in culture. It should be noted that the formulation of GIBCO™ Knockout™ Serum Replacement includes Albumax (Bovine serum albumin enriched with lipids) which is from an animal source (International Patent Publication No. WO 98/30679 to Price, P. J. et al). However, a recent publication by Crook et al., 2007 (Crook J M., et al., 2007, Cell Stem Cell, 1: 490-494) describes six clinical-grade hESC lines generated using FDA-approved clinical grade foreskin fibroblasts in cGMP-manufactured KNOCKOUT™ Serum Replacement (Invitrogen Corporation, USA, Catalogue No. 04-0095).

(38) According to some embodiments of the invention, the concentration of GIBCO™ Knockout KNOCKOUT™ Serum Replacement in the culture medium is in the range of from about 1% [volume/volume (v/v)] to about 50% (v/v), e.g., from about 5% (v/v) to about 40% (v/v), e.g., from about 5% (v/v) to about 30% (v/v), e.g., from about 10% (v/v) to about 30% (v/v), e.g., from about 10% (v/v) to about 25% (v/v), e.g., from about 10% (v/v) to about 20% (v/v), e.g., about 10% (v/v), e.g., about 15% (v/v), e.g., about 20% (v/v), e.g., about 30% (v/v).

(39) Another commercially available serum replacement is the B27 supplement without vitamin A which is available from Gibco-Invitrogen, Corporation, Grand Island, NY USA, Catalogue No. 12587-010. The B27 supplement is a serum-free formulation which includes d-biotin, fatty acid free fraction V bovine serum albumin (BSA), catalase, L-carnitine HCl, corticosterone, ethanolamine HCl, D-galactose (Anhyd.), glutathione (reduced), recombinant human insulin, linoleic acid, linolenic acid, progesterone, putrescine-2-HCl, sodium selenite, superoxide dismutase, T-3/albumin complex, DL alpha-tocopherol and DL alpha tocopherol acetate. However, the use of B27 supplement is limited since it includes albumin from an animal source.

(40) According to some embodiments of the invention, the serum replacement is xeno-free.

(41) The term “xeno” is a prefix based on the Greek word “Xenos”, i.e., a stranger. As used herein the phrase “xeno-free” refers to being devoid of any components which are derived from a xenos (i.e., not the same, a foreigner) species. Such components can be contaminants such as pathogens associated with (e.g., infecting) the xeno species, cellular components of the xeno species or a-

cellular components (e.g., fluid) of the xeno species.

(42) For example, a xeno-free serum replacement can include a combination of insulin, transferrin and selenium. Additionally or alternatively, a xeno-free serum replacement can include human or recombinantly produced albumin, transferrin and insulin.

(43) Non-limiting examples of commercially available xeno-free serum replacement compositions include the premix of ITS (Insulin, Transferrin and Selenium) available from Invitrogen corporation (ITS, Invitrogen, Catalogue No. 5150-056); Serum replacement 3 (Sigma, Catalogue No. S2640) which includes human serum albumin, human transferrin and human recombinant insulin and does not contain growth factors, steroid hormones, glucocorticoids, cell adhesion factors, detectable Ig and mitogens.

(44) According to some embodiments of the invention, the xeno-free serum replacement formulations ITS (Invitrogen corporation) and SR3 (Sigma) are diluted in a 1 to 100 ratio in order to reach a $\times 1$ working concentration.

(45) According to some embodiments of the invention the culture medium is capable of maintaining pluripotent stem cell in a proliferative, pluripotent and undifferentiated state for at least about 5 passages, at least about 10 passages, at least about 15 passages, at least about 20 passages, at least about 22 passages, at least about 25 passages, at least about 30 passages, at least about 35 passages, at least about 40 passages, at least about 45 passages, at least about 50 passages and more.

(46) According to some embodiments of the invention the culture medium is capable of expanding the pluripotent stem cells in an undifferentiated state.

(47) As used herein the term “expanding” refers to increasing the number of pluripotent stem cells over the culturing period (by at least about 5%, 10%, 15%, 20%, 30%, 50%, 100%, 200%, 500%, 1000%, and more). It will be appreciated that the number of pluripotent stem cells which can be obtained from a single pluripotent stem cell depends on the proliferation capacity of the pluripotent stem cell. The proliferation capacity of a pluripotent stem cell can be calculated by the doubling time of the cell (i.e., the time needed for a cell to undergo a mitotic division in the culture) and the period the pluripotent stem cell culture can be maintained in the undifferentiated state (which is equivalent to the number of passages multiplied by the days between each passage).

(48) For example, as described in Example 1 of the Examples section which follows, the hESCs or human iPS cells could be maintained in the proliferative, pluripotent and undifferentiated state in the presence of the mHA40/4, HA75, HA76, HA78 and HA74/1 culture media for at least 22 passages when cultured on a feeder-free matrix. Given that each passage occurs every 4-7 days, the hESCs or human iPS cells were maintained for 110 days (i.e., 2640 hours). Given that the hESCs or human iPS doubling time was 36 hours, a single hESC or human iPS cell cultured under these conditions could be expanded to give rise to 2^{73} (i.e., 9.4×10^{21}) hESCs or human iPS cells.

(49) According to some embodiments of the invention, the culture medium of some embodiments of the invention is capable of supporting expansion of a single pluripotent stem cell (e.g., hESC or human iPS cell) or a population of pluripotent stem cells by at least 2^3 (i.e., 8×10^6) within about one month, e.g., at least 2^{24} (i.e., 16.7×10^6) within about one month.

(50) According to some embodiments of the invention the serum-free and xeno-free culture medium comprises basic fibroblast growth factor (bFGF), transforming growth factor beta-3 (TGF β .3) and ascorbic acid, wherein a concentration of the ascorbic acid in the culture medium is at least 50 μ g/ml and wherein the culture medium is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.

(51) Ascorbic acid (also known as vitamin C) is a sugar acid (C₆H₈O₆; molecular weight 176.12 grams/mole) with antioxidant properties. The ascorbic acid used by the culture medium of some embodiments of the invention can be a natural ascorbic acid, a synthetic ascorbic acid, an ascorbic acid salt (e.g., sodium ascorbate, calcium ascorbate, potassium ascorbate), an ester

form of ascorbic acid (e.g., ascorbyl palmitate, ascorbyl stearate), a functional derivative thereof (a molecule derived from ascorbic acid which exhibits the same activity/function when used in the culture medium of the invention), or an analogue thereof (e.g., a functional equivalent of ascorbic acid which exhibits an activity analogous to that observed for ascorbic acid when used in the culture medium of the invention). Non-limiting examples of ascorbic acid formulations which can be used in the culture medium of some embodiments of the invention include L-ascorbic acid and ascorbic acid 3-phosphate.

(52) Ascorbic acid can be obtained from various manufacturers such as Sigma, St Louis, MO, USA (e.g., Catalogue numbers: A2218, A5960, A7506, A0278, A4403, A4544, A2174, A2343, 95209, 33034, 05878, 95210, 95212, 47863, 01-6730, 01-6739, 255564, A92902, W210901).

(53) As mentioned, the concentration of ascorbic acid in the culture medium is at least about 50 µg/ml. According to some embodiments of the invention, the ascorbic acid can be used in a range of concentrations such as from about 50 µg/ml to about 50 mg/ml, e.g., from about 50 µg/ml to about 5 mg/ml, e.g., from about 50 µg/ml to about 1 mg/ml, e.g., from about 100 µg/ml to about 800 µg/ml, e.g., from about 200 µg/ml to about 800 µg/ml, e.g., from about 300 µg/ml to about 700 µg/ml, e.g., from about 400 µg/ml to about 600 µg/ml, e.g., from about 450 µg/ml to about 550 µg/ml.

(54) According to some embodiments of the invention the concentration of ascorbic acid in the culture medium is at least about 75 µg/ml, e.g., at least about 100 µg/ml, e.g., at least about 150 µg/ml, e.g., at least about 200 µg/ml, e.g., at least about 250 µg/ml, e.g., at least about 300 µg/ml, e.g., at least about 350 µg/ml, e.g., at least about 400 µg/ml, e.g., at least about 450 µg/ml, e.g., about 500 µg/ml.

(55) As is shown in Example 1 of the Examples section which follows, the present inventors have used various culture media which include ascorbic acid at a concentration of at least 50 µg/ml (e.g., the mHA40/4, HA75, HA76, HA77, HA78 and HA74/1 culture media) to successfully culture hESCs and iPS cells and maintain them in a proliferative, pluripotent and undifferentiated state for at least 15 passages in the absence of feeder cell support.

(56) Basic fibroblast growth factor (also known as bFGF, FGF2 or FGF-β) is a member of the fibroblast growth factor family. The bFGF used in the culture medium of some embodiments of the invention can be a purified, a synthetic or a recombinantly expressed bFGF protein [(e.g., human bFGF polypeptide GenBank Accession No. NP_001997.5 (SEQ ID NO:31); human bFGF polynucleotide GenBank Accession No. NM_002006.4 (SEQ ID NO:32). It should be noted that for the preparation of a xeno-free culture medium the bFGF is preferably purified from a human source or is recombinantly expressed as is further described hereinbelow. bFGF can be obtained from various commercial sources such as Cell Sciences®, Canton, MA, USA (e.g., Catalogue number CRF001A and CRF001B), Invitrogen Corporation products, Grand Island NY, USA (e.g., Catalogue numbers: PHG0261, PHG0263, PHG0266 and PHG0264), ProSpec-Tany TechnoGene Ltd. Rehovot, Israel (e.g., Catalogue number: CYT-218), and Sigma, St Louis, MO, USA (e.g., catalogue number: F0291).

(57) According to some embodiments the concentration of bFGF in culture medium is in the range from about 1 ng/ml to about 10 µg/ml, e.g., from about 2 ng/ml to about 1 µg/ml, e.g., from about 1 ng/ml to about 500 ng/ml, e.g., from about 2 ng/ml to about 500 ng/ml, e.g., from about 5 ng/ml to about 250 ng/ml, e.g., from about 5 ng/ml to about 200 ng/ml, e.g., from about 5 ng/ml to about 150 ng/ml, e.g., about 10 ng/ml, e.g., about 20 ng/ml, e.g., about 30 ng/ml, e.g., about 40 ng/ml, e.g., about 50 ng/ml, e.g., about 60 ng/ml, e.g., about 70 ng/ml, e.g., about 80 ng/ml, e.g., about 90 ng/ml, e.g., about 100 ng/ml, e.g., about 110 ng/ml, e.g., about 120 ng/ml, e.g., about 130 ng/ml, e.g., about 140 ng/ml, e.g., about 150 ng/ml.

(58) According to some embodiments of the invention the concentration of bFGF in the culture medium is at least about 1 ng/ml, at least about 2 ng/ml, at least about 3 ng, at least about 4 ng/ml, at least about 5 ng/ml, at least about 6 ng/ml, at least about 7 ng, at least about 8 ng/ml, at least

about 9 ng/ml, at least about 10 ng/ml, at least about 15 ng/ml, at least about 20 ng/ml, at least about 25 ng/ml, at least about 30 ng/ml, at least about 35 ng/ml, at least about 40 ng/ml, at least about 45 ng/ml, at least about 50 ng/ml, at least about 55 ng/ml, at least about 60 ng/ml, at least about 70 ng/ml, at least about 80 ng/ml, at least about 90 ng/ml, at least about 95 ng/ml, e.g., about 100 ng/ml.

(59) As is shown in Example 1 of the Examples section which follows, the present inventors have used various culture media which include bFGF in the range of 5-200 ng/ml (e.g., the mHA40/4, HA75 and HA78 culture media, which include 10 ng/ml bFGF; the HA76 and HA77 culture media which include 100 ng/ml bFGF; and the HA74/1 culture medium which includes 50 ng/ml bFGF) to successfully culture hESCs and iPS cells and maintain them in a proliferative, pluripotent and undifferentiated state for at least 15 passages in the absence of feeder cell support.

(60) Transforming growth factor beta-3 (TGF β .sub.3) is involved in the control of proliferation, differentiation, and other functions in many cell types, acts in inducing transformation and as a negative autocrine growth factor. TGF β 3 can be obtained from various commercial sources such as R&D Systems Minneapolis MN, USA.

(61) According to some embodiments of the invention, the concentration of TGF β .sub.3 in the culture medium is in the range of about 0.05 ng/ml to about 1 μ g/ml, e.g., from 0.1 ng/ml to about 1 μ g/ml, e.g., from about 0.5 ng/ml to about 100 ng/ml.

(62) According to some embodiments of the invention, the concentration of TGF β .sub.3 in the culture medium is at least about 0.5 ng/ml, e.g., at least about 0.6 ng/ml, e.g., at least about 0.8 ng/ml, e.g., at least about 0.9 ng/ml, e.g., at least about 1 ng/ml, e.g., at least about 1.2 ng/ml, e.g., at least about 1.4 ng/ml, e.g., at least about 1.6 ng/ml, e.g., at least about 1.8 ng/ml, e.g., about 2 ng/ml.

(63) As is shown in Example 1 of the Examples section which follows, the present inventors have used various culture media which include TGF β .sub.3 at a concentration of about 2 ng/ml (e.g., the mHA40/4, HA75, HA76, HA78 and HA74/1 culture media) to successfully culture hESCs and iPS cells and maintain them in a proliferative, pluripotent and undifferentiated state for at least 22 passages in the absence of feeder cell support.

(64) According to some embodiments of the invention, the culture medium comprises bFGF at a concentration range of about 0.1 ng/ml to about 500 ng/ml, TGF β 3 at a concentration range of about 0.1 ng/ml to about 20 ng/ml, and ascorbic acid at a concentration range of about 50 μ g/ml to about 5000 μ g/ml.

(65) According to some embodiments of the invention, the culture medium of some embodiments of the invention comprises bFGF at a concentration range of about 5 ng/ml to about 150 ng/ml, TGF β 3 at a concentration range of about 0.5 ng/ml to about 5 ng/ml, and ascorbic acid at a concentration range of about 400 μ g/ml to about 600 μ g/ml.

(66) According to some embodiments of the invention, the culture medium further comprises a lipid mixture.

(67) As used herein the phrase "lipid mixture" refers to a defined (e.g., chemically defined) lipid composition needed for culturing the pluripotent stem cells. It should be noted that the lipid mixture is usually added to a culture medium which is devoid of serum or serum replacement and thus substitutes the lipids which are usually added to formulations of serum or serum replacement.

(68) A non-limiting example of a commercially available lipid mixture, which can be used in the culture medium of some embodiments of the invention, include the Chemically Define Lipid Concentrate available from Invitrogen (Catalogue No. 11905-031).

(69) According to some embodiments of the invention, the concentration of the lipid mixture in the culture medium is from about 0.5% [volume/volume (v/v)] to about 3% v/v, e.g., from about 0.5% v/v to about 2% v/v, e.g., from about 0.5% v/v to about 1% v/v, e.g., about 1% v/v.

(70) According to some embodiments of the invention, the culture medium of some embodiments of the invention comprises bFGF at a concentration range of about 0.1 ng/ml to about 500 ng/ml,

TGFβ.sub.3 at a concentration range of about 0.1 ng/ml to about 20 ng/ml, ascorbic acid at a concentration range of about 50 μg/ml to about 5000 μg/ml, xeno-free serum replacement and a lipid mixture.

(71) Non-limiting examples of xeno-free and serum-free culture media which comprise TGFβ.sub.3, bFGF and ascorbic acid at a concentration of at least 50 μg/ml and which can be used to maintain pluripotent stem cells in a proliferative and undifferentiated states include the HA75 and HA78 culture media.

(72) According to some embodiments of the invention, the culture medium further comprises sodium bicarbonate. Sodium bicarbonate can be obtained from Biological Industries, Beit HaEmek, Israel.

(73) According to some embodiments of the invention, the concentration of sodium bicarbonate in the culture medium is from about 5% to about 10%, e.g., from about 6% to about 9%, e.g., from about 7% to about 8%, e.g., about 7.5%.

(74) The present inventors uncovered that pluripotent stem cells can be maintained in a proliferative, pluripotent and undifferentiated state for at least 15 passages when cultured in a serum-free and xeno-free culture medium which comprises bFGF and ascorbic acid but does not comprise a TGFβ isoform.

(75) As used herein the phrase "TGFβ isoform" refers to any isoform of the transforming growth factor beta (β) including TGFβ.sub.1 (e.g., *Homo sapiens* TGFβ.sub.1, GenBank Accession No. NP_000651), TGFβ.sub.2 (e.g., *Homo sapiens* TGFβ.sub.2, GenBank Accession No. NP_003229) and TGFβ3 (e.g., *Homo sapiens* TGFβ.sub.3, GenBank Accession No. NP_003230) which functions through the same receptor signaling system in the control of proliferation, differentiation, and other functions in many cell types. TGFβ acts in inducing transformation and also acts as a negative autocrine growth factor.

(76) According to some embodiments of the invention, the culture medium comprises no more than 1 ng/ml of the TGFβ isoform, e.g., no more than 0.5 ng/ml, e.g., no more than 0.1 ng/ml, e.g., no more than 0.05 ng/ml, e.g., no more than 0.01 ng/ml of the TGFβ isoform.

(77) According to some embodiments of the invention, the culture medium is completely devoid of a TGFβ isoform (i.e., TGFβ isoform-free).

(78) According to some embodiments of the invention the culture medium comprises ascorbic acid at a concentration range of about 400-600 μg/ml and basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml.

(79) According to some embodiments of the invention the culture medium the culture medium which comprises ascorbic acid at a concentration range of about 400-600 μg/ml and basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml is capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support.

(80) According to some embodiments of the invention, the concentration of ascorbic acid in the culture medium is between about 410 μg/ml to about 590 μg/ml, between about 420 μg/ml to about 580 μg/ml, between about 450 μg/ml to about 550 μg/ml, between about 460 μg/ml to about 540 μg/ml, between about 470 μg/ml to about 530 μg/ml, between about 490 μg/ml to about 520 μg/ml, e.g., between about 490 μg/ml to about 510 μg/ml, e.g., about 500 μg/ml.

(81) According to some embodiments of the invention, the concentration of bFGF in the culture medium is between about 50 ng/ml to about 200 ng/ml, between about 60 ng/ml to about 190 ng/ml, between about 70 ng/ml to about 180 ng/ml, between about 80 ng/ml to about 170 ng/ml, between about 90 ng/ml to about 160 ng/ml, between about 90 ng/ml to about 150 ng/ml, between about 90 ng/ml to about 130 ng/ml, between about 90 ng/ml to about 120 ng/ml, e.g., about 100 ng/ml.

(82) According to some embodiments of the invention, the concentration of bFGF in the culture medium is about 50, about 55, about 60, about 65, about 70, about 80, about 85, about 90, about 95, about 100, about 105, about 110, about 115, about 120, about 125, about 130, about 135, about

140, about 145, about 150, about 160, about 165, about 170, about 175, about 180, about 185, about 190, about 195, about 200 ng/ml.

(83) According to some embodiments of the invention the culture medium which comprises ascorbic acid at a concentration range of about 400-600 µg/ml and basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, further comprises xeno-free serum replacement.

(84) According to some embodiments of the invention, the culture medium which comprises ascorbic acid at a concentration range of about 400-600 µg/ml and basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, further comprises a lipid mixture.

(85) According to some embodiments of the invention, the culture medium comprises bFGF at a concentration of about 50-200 ng/ml and ascorbic acid at a concentration of about 400-600 µg/ml is devoid of sodium-bicarbonate.

(86) According to some embodiments of the invention, the culture medium comprises bFGF at a concentration of about 50-200 ng/ml and ascorbic acid at a concentration of about 400-600 µg/ml, xeno-free serum replacement at a concentration of about 1% and lipid mixture at a concentration of about 1%.

(87) A non-limiting example of a xeno-free, serum-free, and TGFβ isoform-free culture medium which comprises ascorbic acid at a concentration range of about 400-600 µg/ml, bFGF at a concentration range of about 50-200 ng/ml, xeno-free serum replacement and a lipid mixture and which is capable of maintaining pluripotent stem cells such as hESCs and human iPS cells in a proliferative and undifferentiated state for at least 21 passages in the absence of feeder cell support is the HA77 culture medium (Example 1 of the Examples section which follows) or a culture medium similar to the HA77 medium but which is devoid of sodium bi-carbonate such as a culture medium which consists of DMEM/F12 (94%) (Biological Industries, Israel, Sigma Israel), L-glutamine 2 mM (Invitrogen corporation, Sigma, Israel), ascorbic acid 500 µg/ml (Sigma, Israel), bFGF—100 ng (Invitrogen corporation), SR3—1% (Sigma, Israel), and defined lipid mixture 1% (Invitrogen corporation, Sigma, Israel). The present inventors have uncovered novel serum-free and highly defined culture media, which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional (i.e., a suspension culture) systems in the absence of feeder cell support.

(88) As used herein the phrase “suspension culture” refers to a culture in which the pluripotent stem cells are suspended in a medium rather than adhering to a surface.

(89) According to some embodiments of the invention the serum-free culture medium which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional culture systems in the absence of feeder cell support comprises basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml.

(90) According to some embodiments of the invention the culture medium comprises between about 55-190 ng/ml, e.g., between about 60-190 ng/ml, e.g., between about 70-180 ng/ml, e.g., between about 80-160 ng/ml, e.g., between about 90-150 ng/ml, e.g., between about 90-140 ng/ml, e.g., between about 90-130 ng/ml, e.g., between about 90-120 ng/ml, e.g., between about 90-110 ng/ml, e.g., between about 95-105 ng/ml, e.g., about 100 ng/ml.

(91) According to some embodiments of the invention the culture medium which comprises bFGF between about 50-200 ng/ml further comprises serum replacement.

(92) A non-limiting example of a culture medium which comprises bFGF at a concentration between about 50-200 ng/ml is the YF100 medium which comprises a basic medium (e.g., DMEM/F12, 85%), serum replacement (15%), bFGF (100 ng/ml), L-glutamine (2 mM), β-mercaptoethanol (0.1 mM) and non-essential amino acid stock (1%).

(93) According to some embodiments of the invention the serum-free culture medium which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional culture systems in the absence of feeder cell support consists of a basic medium, ascorbic acid at a concentration range of about 50 µg/ml to about 500 µg/ml,

bFGF at a concentration range between about 2 ng/ml to about 20 ng/ml, L-glutamine, and serum replacement.

(94) According to some embodiments of the invention the serum-free culture medium which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional culture systems in the absence of feeder cell support consists of a basic medium, ascorbic acid at a concentration range of about 50 µg/ml to about 500 µg/ml, bFGF at a concentration range between about 2 ng/ml to about 20 ng/ml, L-glutamine, serum replacement and a lipid mixture.

(95) According to some embodiments of the invention the concentration of ascorbic acid is about 50 µg/ml.

(96) According to some embodiments of the invention the concentration of ascorbic acid is about 500 µg/ml.

(97) According to some embodiments of the invention the concentration of bFGF is about 4 ng/ml.

(98) The basic medium can be any known tissue culture medium such as DMEM/F12 (Biological Industries, Israel, or Sigma Israel), Ko-DMEM (Invitrogen). The concentration of the basic medium depends on the concentration of the other medium ingredients such as the serum replacement.

(99) The serum replacement can be any xeno-free serum replacement (devoid of animal contaminants) at a concentration range from 1-20% depending on the serum replacement used. For example, if the SR3 serum replacement is used then its concentration in the medium is about 1%.

(100) According to some embodiments of the invention the concentration of L-glutamine is about 2 mM.

(101) According to some embodiments of the invention the concentration of the lipid mixture (Sigma, Israel; or Invitrogen, Israel) is about 1%.

(102) Non-limiting examples of such a culture medium include the modified HA13(a) medium [DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 500 µg/ml, bFGF—4 ng, and SR3—1%]; the modified HA13(b) medium [DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 500 µg/ml, bFGF—4 ng, SR3—1% and a lipid mixture (1%)]; the modified HA13(c) medium [DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 50 µg/ml, bFGF—4 ng, and SR3—1%]; and the modified HA13(d) medium [DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 50 µg/ml, bFGF—4 ng, SR3—1% and a lipid mixture (1%)]. These culture media were capable of maintaining pluripotent stem cells (e.g., hESCs and hiPS cells) in a proliferative, pluripotent and undifferentiated state for at least 20 passages when cultured in a two-dimensional (e.g., on a feeder-layer free culture system; data not shown) and for at least 20 passages when cultured on a three-dimensional culture system (e.g., suspension culture without adherence to an external substrate, cell encapsulation or to protein carrier; data not shown).

(103) According to some embodiments of the invention the serum-free culture medium which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional culture systems in the absence of feeder cell support comprises an IL6RIL6 chimera at a concentration range of about 50-200 picogram per milliliter (pg/ml).

(104) As used herein the phrase “IL6RIL6 chimera” refers to a chimeric polypeptide which comprises the soluble portion of interleukin-6 receptor [IL-6-R, e.g., the human IL-6-R as set forth by GenBank Accession No. AAH89410; SEQ ID NO:33; e.g., a portion of the soluble IL6 receptors as set forth by amino acids 112-355 (SEQ ID NO:39) of GenBank Accession No. AAH89410] and the interleukin-6 (IL6; e.g., human IL-6 as set forth by GenBank Accession No. CAG29292; SEQ ID NO:35) or a biologically active fraction thereof (e.g., a receptor binding domain).

(105) It should be noted that when constructing the IL6RIL6 chimera the two functional portions (i.e., the IL6 and its receptor) can be directly fused (e.g., attached or translationally fused, i.e., encoded by a single open reading frame) to each other or conjugated (attached or translationally

fused) via a suitable linker (e.g., a polypeptide linker). According to some embodiments of the invention, the IL6RIL6 chimeric polypeptide exhibits a similar amount and pattern of glycosylation as the naturally occurring IL6 and IL6 receptor. For example, a suitable IL6RIL6 chimera is as set forth in SEQ ID NO:36 and in FIG. 11 of WO 99/02552 to Revel M., et al., which is fully incorporated herein by reference.

(106) It will be appreciated that any of the proteinaceous factors used in the culture medium of the present invention (e.g., the IL6RIL6 chimera, bFGF, TGF β .sub.3) can be recombinantly expressed or biochemically synthesized. In addition, naturally occurring proteinaceous factors such as bFGF and TGF β can be purified from biological samples (e.g., from human serum, cell cultures) using methods well known in the art.

(107) Biochemical synthesis of the proteinaceous factors of the present invention (e.g., the IL6RIL6 chimera) can be performed using standard solid phase techniques. These methods include exclusive solid phase synthesis, partial solid phase synthesis methods, fragment condensation and classical solution synthesis.

(108) Recombinant expression of the proteinaceous factors of the present invention (e.g., the IL6RIL6 chimera) can be generated using recombinant techniques such as described by Bitter et al., (1987) *Methods in Enzymol.* 153:516-544, Studier et al. (1990) *Methods in Enzymol.* 185:60-89, Brisson et al. (1984) *Nature* 310:511-514, Takamatsu et al. (1987) *EMBO J.* 6:307-311, Coruzzi et al. (1984) *EMBO J.* 3:1671-1680, Brogli et al., (1984) *Science* 224:838-843, Gurley et al. (1986) *Mol. Cell. Biol.* 6:559-565 and Weissbach & Weissbach, 1988. *Methods for Plant Molecular Biology*, Academic Press, NY, Section VIII, pp 421-463. Specifically, the IL6RIL6 chimera can be generated as described in PCT publication WO 99/02552 to Revel M., et al. and Chebath J., et al., 1997, which are fully incorporated herein by reference.

(109) According to some embodiments of the invention, the concentration of the IL6RIL6 chimera in the culture medium is in the range from about 55 pg/ml to about 195 pg/ml, e.g., from about 60 pg/ml to about 190 pg/ml, e.g., from about 65 pg/ml to about 185 pg/ml, e.g., from about 70 pg/ml to about 180 pg/ml, e.g., from about 75 pg/ml to about 175 pg/ml, e.g., from about 80 pg/ml to about 170 pg/ml, e.g., from about 85 pg/ml to about 165 pg/ml, e.g., from about 90 pg/ml to about 150 pg/ml, e.g., from about 90 pg/ml to about 140 pg/ml, e.g., from about 90 pg/ml to about 130 pg/ml, e.g., from about 90 pg/ml to about 120 pg/ml, e.g., from about 90 pg/ml to about 110 pg/ml, e.g., from about 95 pg/ml to about 105 pg/ml, e.g., from about 98 pg/ml to about 102 pg/ml, e.g., about 100 pg/ml.

(110) According to some embodiments of the invention, the IL6RIL6 chimera-containing culture medium further comprises bFGF.

(111) According to some embodiments of the invention, concentration of bFGF in the IL6RIL6 chimera-containing culture medium is in the range of from about 1 ng/ml to about 10 μ g/ml, e.g., from about 2 ng/ml to about 1 μ g/ml, e.g., from about 2 ng/ml to about 500 ng/ml, e.g., from about 5 ng/ml to about 150 ng/ml, e.g., from about 5 ng/ml to about 100 ng/ml, e.g., from about 5 ng/ml to about 80 ng/ml, e.g., from about 5 ng/ml to about 50 ng/ml, e.g., from about 5 ng/ml to about 30 ng/ml, e.g., about 5 ng/ml, e.g., about 10 ng/ml, e.g., about 15 ng/ml, e.g., about 20 ng/ml.

(112) According to some embodiments of the invention, the IL6RIL6 chimera-containing culture medium further comprises serum replacement.

(113) According to some embodiments of the invention, the concentration of KNOCKOUT™ Serum Replacement in the IL6RIL6 chimera-containing culture medium is in the range from about 1% (v/v) to about 50% (v/v), e.g., from about 5% (v/v) to about 40% (v/v), e.g., from about 5% (v/v) to about 30% (v/v), e.g., from about 10% (v/v) to about 30% (v/v), e.g., from about 10% (v/v) to about 25% (v/v), e.g., from about 10% (v/v) to about 20% (v/v), e.g., about 15% (v/v).

(114) According to some embodiments of the invention, the culture medium comprises IL6RIL6 chimera at a concentration range of about 50-200 pg/ml, bFGF at a concentration range of about 5-50 ng/ml and serum replacement at a concentration of about 5-40%.

(115) For example, as is shown in Example 4 of the Examples section which follows, the CM100Fp culture medium was shown capable of maintaining pluripotent stem cells such as hESCs and human iPS cells in a proliferative, pluripotent and undifferentiated state for at least 50 passages in a suspension culture devoid of substrate adherence.

(116) According to some embodiments of the invention, the serum-free culture medium which can maintain pluripotent stem cells in a proliferative, pluripotent and undifferentiated state in two-dimensional and three-dimensional culture systems in the absence of feeder-cells support comprises LIF at a concentration of at least 2000 units/ml.

(117) Leukemia inhibitory factor (LIF) is a pleiotropic cytokine which is involved in the induction of hematopoietic differentiation, induction of neuronal cell differentiation, regulator of mesenchymal to epithelial conversion during kidney development, and may also have a role in immune tolerance at the maternal-fetal interface. The LIF used in the culture medium of some embodiments of the invention can be a purified, synthetic or recombinantly expressed LIF protein [e.g., human LIF polypeptide GenBank Accession No. NP_002300.1 (SEQ ID NO:37); human LIF polynucleotide GenBank Accession No. NM_002309.3 (SEQ ID NO:38)]. It should be noted that for the preparation of a xeno-free culture medium LIF is preferably purified from a human source or is recombinantly expressed. Recombinant human LIF can be obtained from various sources such as Chemicon, USA (Catalogue No. LIF10100) and AbD Serotec (MorphoSys US Inc, Raleigh, NC 27604, USA). Murine LIF ESGRO® (LIF) can be obtained from Millipore, USA (Catalogue No. ESG1107).

(118) According to some embodiments of the invention, the concentration of LIF in the culture medium is from about 2000 units/ml to about 10,000 units/ml, e.g., from about 2000 units/ml to about 8,000 units/ml, e.g., from about 2000 units/ml to about 6,000 units/ml, e.g., from about 2000 units/ml to about 5,000 units/ml, e.g., from about 2000 units/ml to about 4,000 units/ml.

(119) According to some embodiments of the invention, the concentration of LIF in the culture medium is at least about 2000 units/ml, e.g., at least about 2100 units/ml, e.g., at least about 2200 units/ml, e.g., at least about 2300 units/ml, e.g., at least about 2400 units/ml, e.g., at least about 2500 units/ml, e.g., at least about 2600 units/ml, e.g., at least about 2700 units/ml, e.g., at least about 2800 units/ml, e.g., at least about 2900 units/ml, e.g., at least about 2950 units/ml, e.g., about 3000 units/ml.

(120) According to some embodiments of the invention, the LIF-containing culture medium further comprises bFGF.

(121) The concentration of bFGF in the LIF-containing culture medium is in the range of about 0.1 ng/ml to about 10 µg/ml, e.g., from about 2 ng/ml to about 1 µg/ml, e.g., from about 2 ng/ml to about 500 ng/ml, e.g., from about 5 ng/ml to about 150 ng/ml, e.g., from about 5 ng/ml to about 100 ng/ml, e.g., from about 5 ng/ml to about 80 ng/ml, e.g., from about 5 ng/ml to about 50 ng/ml, e.g., from about 5 ng/ml to about 30 ng/ml, e.g., about 5 ng/ml, e.g., about 10 ng/ml, e.g., about 15 ng/ml, e.g., about 20 ng/ml.

(122) According to some embodiments of the invention, the LIF-containing culture medium further comprises serum replacement.

(123) According to some embodiments of the invention, the culture medium comprises LIF at a concentration of about 2000-10,000 units/ml, bFGF at a concentration range from about 0.1 ng/ml to about 10 µg/ml and KNOCKOUT™ Serum Replacement at a concentration range from about 1% (v/v) to about 50% (v/v).

(124) According to some embodiments of the invention, the culture medium comprises LIF at a concentration of about 2000-5,000 units/ml, bFGF at a concentration of about 5-50 ng/ml and serum replacement at a concentration of about 5-30%.

(125) For example, as shown in Example 4 of the Examples section which follows, the yFL3 culture medium was shown capable of maintaining pluripotent stem cells such as human ESCs and human iPS cells in a proliferative, pluripotent and undifferentiated state for at least 10 passages

when cultured in a suspension culture.

(126) According to some embodiments of the invention, the ingredients included in the culture medium of some embodiments of the invention are substantially pure, with a tissue culture and/or a clinical grade.

(127) According to an aspect of some embodiments of the invention there is provided a cell culture which comprises the pluripotent stem cell of some embodiments of the invention and the culture medium of some embodiments of the invention.

(128) According to an aspect of some embodiments of the invention cell culture is feeder cells free (e.g., being devoid of feeder cells or feeder cell conditioned medium).

(129) According to some embodiments of the invention the pluripotent stem cells which are included in the cell culture of some embodiments of the invention exhibit a stable karyotype (chromosomal stability) during the culturing period, e.g., for at least 2 passages, e.g., at least 4 passages, e.g., at least 8 passages, e.g., at least 15 passages, e.g., at least 20 passages, e.g., at least 25 passages, e.g., at least 30 passages, e.g., at least 35 passages, e.g., at least 40 passages, e.g., at least 45 passages, e.g., at least 50 passages.

(130) According to some embodiments of the invention, the cell culture of the invention exhibit a doubling time of at least 20 hours, e.g., a doubling time which is between 20 to 40 hours (e.g., about 36 hours), thus representing a non-tumorigenic, genetically stable pluripotent stem cells (e.g., hESCs and iPS cells).

(131) According to some embodiments of the invention, the cell culture of the invention is characterized by at least 40%, at least 50%, at least 60%, e.g., at least 70%, e.g., at least 80%, e.g., at least 85%, e.g., at least 90%, e.g., at least 95% of undifferentiated pluripotent stem cells.

(132) According to an aspect of some embodiments of the invention, there is provided a method of expanding and maintaining pluripotent stem cells in a pluripotent and undifferentiated state.

(133) According to some embodiments of the invention, the method of expanding and maintaining pluripotent stem cells in an undifferentiated state is effected by culturing the pluripotent stem cells in any of the novel culture media of the invention (described herein).

(134) According to some embodiments of the invention, the method of expanding and maintaining pluripotent stem cells in an undifferentiated state is effected by culturing the pluripotent stem cells in a culture medium being serum-free, feeder-free, matrix-free and protein carrier-free and comprising basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml.

(135) According to some embodiments of the invention culturing is effected on a two-dimensional culture system such as a matrix or a feeder cell layer.

(136) For example, culturing on a two-dimensional culture system can be performed by plating the pluripotent stem cells onto a matrix or a feeder cell layer in a cell density which promotes cell survival and proliferation but limits differentiation. Typically, a plating density of between about 15,000 cells/cm.² and about 3,000,000 cells/cm.² is used.

(137) It will be appreciated that although single-cell suspensions of pluripotent stem cells are usually seeded, small clusters may also be used. To this end, enzymatic digestion (such as with type IV collagenase) utilized for cluster disruption (see "General Materials and Experimental Methods" in the Examples section which follows) is terminated before stem cells become completely dispersed and the cells are triturated with a pipette such that clumps (i.e., 10-200 cells) are formed. However, measures are taken to avoid large clusters which may cause cell differentiation.

(138) As used herein, the term "matrix" refers to any substance to which the pluripotent stem cells can adhere and which therefore can substitute the cell attachment function of feeder cells. Such a matrix typically contains extracellular components to which the pluripotent stem cells can attach and thus it provides a suitable culture substrate.

(139) According to some embodiments of the invention the matrix comprises an extracellular matrix.

(140) The extracellular matrix can be composed of components derived from basement membrane

or extracellular matrix components that form part of adhesion molecule receptor-ligand couplings. MATRIGEL® (Becton Dickinson, USA) is one example of a commercially available matrix which is suitable for use with the present invention. MATRIGEL® is a soluble preparation from Engelbreth-Holm-Swarm tumor cells that gels at room temperature to form a reconstituted basement membrane; MATRIGEL® is also available as a growth factor reduced preparation. Other extracellular matrix components and component mixtures which are suitable for use with the present invention include foreskin matrix, laminin matrix, fibronectin matrix, proteoglycan matrix, entactin matrix, heparan sulfate matrix, collagen matrix and the like, alone or in various combinations thereof.

(141) According to some embodiments of the invention the matrix is xeno-free.

(142) In cases where complete animal-free culturing conditions are desired, the matrix is preferably derived from a human source or synthesized using recombinant techniques such as described hereinabove. Such matrices include, for example, human-derived fibronectin, recombinant fibronectin, human-derived laminin, foreskin fibroblast matrix or a synthetic fibronectin matrix. Human derived fibronectin can be from plasma fibronectin or cellular fibronectin, both of which can be obtained from Sigma, St. Louis, MO, USA. Human derived laminin and foreskin fibroblast matrix can be obtained from Sigma, St. Louis, MO, USA. A synthetic fibronectin matrix can be obtained from Sigma, St. Louis, MO, USA.

(143) According to some embodiments of the invention, culturing is effected on a feeder cell layer.

(144) According to some embodiments of the invention, the method of expanding and maintaining pluripotent stem cells in an undifferentiated state is effected by culturing the pluripotent stem cells on a feeder cell layer in a serum-free and xeno-free culture medium which comprises basic fibroblast growth factor (bFGF), transforming growth factor beta-3 (TGFβ3) and ascorbic acid, wherein a concentration of the ascorbic acid in the culture medium is at least 50 µg/ml.

(145) According to some embodiments of the invention, the method of expanding and maintaining pluripotent stem cells in an undifferentiated state is effected by culturing the pluripotent stem cells on a feeder cell layer in a serum-free and xeno-free culture medium which comprises ascorbic acid at a concentration range of about 400-600 µg/ml, basic fibroblast growth factor (bFGF) at a concentration range of about 50-200 ng/ml, xeno-free serum replacement and a lipid mixture.

(146) According to some embodiments of the invention, the feeder cell layer is xeno-free.

(147) According to some embodiments of the invention, the feeder cell layer is a foreskin fibroblasts feeder cell layer.

(148) According to some embodiments of the invention, culturing according to some embodiments of the invention is effected in a suspension culture.

(149) According to some embodiments of the invention, the suspension culture is devoid of substrate adherence, e.g., without adherence to an external substrate such as components of extracellular matrix, a glass microcarrier or beads.

(150) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in a protein carrier-free culture medium.

(151) As used herein the phrase “protein carrier” refers to a protein which acts in the transfer of proteins or nutrients (e.g., minerals such as zinc) to the cells in the culture. Such protein carriers can be, for example, albumin (e.g., bovine serum albumin), Albumax (lipid enriched albumin) or plasmanate (human plasma isolated proteins). Since these carriers are derived from either human or animal sources their use in hESCs of human iPS cell cultures is limited by batch-specific variations and/or exposure to pathogens. Thus, a culture medium which is devoid of a protein carrier (e.g., albumin) is highly advantageous since it enables a truly defined medium that can be manufactured from recombinant or synthetic materials.

(152) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in a serum-free and feeder cell-free culture medium.

(153) It should be noted that some protocols of culturing pluripotent stem cells such as hESCs and

iPS cells include microencapsulation of the cells inside a semipermeable hydrogel membrane, which allows the exchange of nutrients, gases, and metabolic products with the bulk medium surrounding the capsule (for details see e.g., U.S. Patent Application No. 20090029462 to Beardsley et al.).

(154) According to some embodiments of the invention, the pluripotent stem cells cultured in the suspension culture are devoid of cell encapsulation.

(155) According to an aspect of some embodiments of the invention, there is provided a method of expanding induced pluripotent stem (iPS) cells and maintaining the iPS cells in an undifferentiated state. The method is effected by culturing the iPS cells in a suspension culture under culturing conditions devoid of substrate adherence and devoid of cell encapsulation and which allow expansion of the iPS cells in the undifferentiated state.

(156) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in the presence of the IL6RIL6 chimera-containing culture medium in which the concentration of the IL6RIL6 chimera is in the range of about 50-200 picograms per milliliter (pg/ml).

(157) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in the presence of the leukemia inhibitory factor (LIF)-containing culture medium in which the concentration of LIF is at least about 2000 units/ml.

(158) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in the presence of a medium which comprises basic fibroblast growth factor (bFGF) at a concentration range of about 50 ng/ml to about 200 ng/ml, e.g., between about 60 ng/ml to about 190 ng/ml, e.g., between about 70 ng/ml to about 180 ng/ml, e.g., between about 80 ng/ml to about 170 ng/ml, e.g., between about 90 ng/ml to about 160 ng/ml, e.g., between about 90 ng/ml to about 150 ng/ml, e.g., between about 90 ng/ml to about 130 ng/ml, e.g., between about 90 ng/ml to about 120 ng/ml, e.g., about 100 ng/ml.

(159) For example, a non-limiting example of a medium which was found suitable for culturing hESCs and human iPS cells in a suspension culture devoid of substrate adherence and cell encapsulation is the yF100 medium which comprises serum replacement and 100 ng/ml bFGF.

(160) According to some embodiments of the invention, culturing of the pluripotent stem cells in a suspension culture is effected in the presence of a medium which comprises the IL6RIL6 chimera at a concentration range of about 50-200 nanogram per milliliter (ng/ml) and bFGF at a concentration in the range of 1-50 ng/ml.

(161) For example, a non-limiting example of a medium which was found suitable for culturing hESCs and human iPS cells in a suspension culture devoid of substrate adherence and cell encapsulation is the CM100F medium which comprises serum replacement, the IL6RIL6 chimera at a concentration of 100 ng/ml and bFGF at a concentration of 10 ng/ml.

(162) For example, using the CM100Fp, CM100F, yF100 or yFL3 culture media the present inventors expanded pluripotent stem cells in a suspension culture in a proliferative, pluripotent and undifferentiated state for at least 50 passages (see e.g., FIGS. 3A-C, 4, 5A-C and 6A-D and is described in Examples 4 and 5 of the Examples section which follows).

(163) Culturing in a suspension culture according to the method of some embodiments of the invention is effected by plating the pluripotent stem cells in a culture vessel at a cell density which promotes cell survival and proliferation but limits differentiation. Typically, a plating density of between about 5×10^4 to 2×10^6 cells per ml is used. It will be appreciated that although single-cell suspensions of stem cells are usually seeded, small clusters such as 10-200 cells may also be used.

(164) In order to provide the pluripotent stem cells with sufficient and constant supply of nutrients and growth factors while in the suspension culture, the culture medium can be replaced on a daily basis, or, at a pre-determined schedule such as every 2-3 days. For example, replacement of the culture medium can be performed by subjecting the pluripotent stem cells suspension culture to

centrifugation for about 3 minutes at 80 g, and resuspension of the formed pluripotent stem cells pellet in a fresh medium. Additionally or alternatively, a culture system in which the culture medium is subject to constant filtration or dialysis so as to provide a constant supply of nutrients or growth factors to the pluripotent stem cells may be employed.

(165) Since large clusters of pluripotent stem cells may cause cell differentiation, measures are taken to avoid large pluripotent stem cells aggregates. According to some embodiments of the invention, the formed pluripotent stem cells clumps are dissociated every 5-7 days and the single cells or small clumps of cells are either split into additional culture vessels (i.e., passaged) or remained in the same culture vessel yet with additional culture medium. For dissociation of large pluripotent stem cells clumps, a pellet of pluripotent stem cells (which may be achieved by centrifugation as described hereinabove) or an isolated pluripotent stem cells clump can be subject to enzymatic digestion and/or mechanical dissociation.

(166) Enzymatic digestion of pluripotent stem cells clump(s) can be performed by subjecting the clump(s) to an enzyme such as type IV Collagenase (Worthington biochemical corporation, Lakewood, NJ, USA) and/or Dispase (Invitrogen Corporation products, Grand Island NY, USA). The time of incubation with the enzyme depends on the size of cell clumps present in the suspension culture. Typically, when pluripotent stem cells cell clumps are dissociated every 5-7 days while in the suspension culture, incubation of 20-60 minutes with 1.5 mg/ml type IV Collagenase results in small cell clumps which can be further cultured in the undifferentiated state. Alternatively, pluripotent stem cells clumps can be subjected to incubation of about 25 minutes with 1.5 mg/ml type IV Collagenase followed by five minutes incubation with 1 mg/ml Dispase. It should be noted that passaging of human ESCs with trypsin may result in chromosomal instability and abnormalities (see for example, Mitalipova M M., et al., Nature Biotechnology, 23: 19-20, 2005 and Cowan C A et al., N. Engl. J. of Med. 350: 1353-1356, 2004). According to some embodiments of the invention, passaging hESC or iPS cell with trypsin should be avoided.

(167) Mechanical dissociation of large pluripotent stem cells clumps can be performed using a device designed to break the clumps to a predetermined size. Such a device can be obtained from CellArtis Goteborg, Sweden. Additionally or alternatively, mechanical dissociation can be manually performed using a needle such as a 27 g needle (BD Microlance, Drogheda, Ireland) while viewing the clumps under an inverted microscope.

(168) According to some embodiments of the invention, following enzymatic or mechanical dissociation of the large cell clumps, the dissociated pluripotent stem cells clumps are further broken to small clumps using 200 μ l Gilson pipette tips (e.g., by pipetting up and down the cells).

(169) The culture vessel used for culturing the pluripotent stem cells in suspension according to the method of some embodiments of the invention can be any tissue culture vessel (e.g., with a purity grade suitable for culturing pluripotent stem cells) having an internal surface designed such that pluripotent stem cells cultured therein are unable to adhere or attach to such a surface (e.g., non-tissue culture treated cells, to prevent attachment or adherence to the surface). Preferably, in order to obtain a scalable culture, culturing according to some embodiments of the invention is effected using a controlled culturing system (preferably a computer-controlled culturing system) in which culture parameters such as temperature, agitation, pH, and pO₂ is automatically performed using a suitable device. Once the culture parameters are recorded, the system is set for automatic adjustment of culture parameters as needed for pluripotent stem cells expansion.

(170) As described in the Examples section which follows, the pluripotent stem cells were cultured under dynamic conditions (i.e., under conditions in which the pluripotent stem cells are subject to constant movement while in the suspension culture; see e.g., FIGS. 6A-D; Example 5) or under non-dynamic conditions (i.e., a static culture; see e.g., FIGS. 3A-C, 4 and 5A-C; Example 4) while preserving their, proliferative, pluripotent capacity and karyotype stability for at least 30 passages.

(171) For non-dynamic culturing of pluripotent stem cells, the pluripotent stem cells can be cultured in uncoated 58 mm Petri dishes (Greiner, Frickenhausen, Germany).

(172) For dynamic culturing of pluripotent stem cells, the pluripotent stem cells can be cultured in spinner flasks [e.g., of 200 ml to 1000 ml, for example 250 ml which can be obtained from CellSpin of Integra Biosciences, Fernwald, Germany; of 100 ml which can be obtained from Bellco, Vineland, NJ; or in 125 ml Erlenmeyer (Corning Incorporated, Corning NY, USA)] which can be connected to a control unit and thus present a controlled culturing system. The culture vessel (e.g., a spinner flask, an Erlenmeyer) is shaken continuously. According to some embodiments of the invention the culture vessels are shaken at 90 rounds per minute (rpm) using a shaker (S3.02.10L, ELMI ltd, Riga, Latvia). According to some embodiments of the invention the culture medium is changed daily.

(173) According to some embodiments of the invention, when cultured according to the teachings of the present invention, the growth of the pluripotent stem cells is monitored to determine their differentiation state. The differentiation state can be determined using various approaches including, for example, morphological evaluation (e.g., as shown in FIGS. 1A-C, 3A-C) and/or detection of the expression pattern of typical markers of the undifferentiated state using immunological techniques such as flow cytometry for membrane-bound markers, immunohistochemistry or immunofluorescence for extracellular and intracellular markers and enzymatic immunoassay, for secreted molecular markers. For example, immunofluorescence employed on hESCs or human iPS cells cultured according to the method of some embodiments of the invention revealed the expression of Oct4, stage-specific embryonic antigen (SSEA) 4, the tumor-rejecting antigen (TRA)-1-60 and TRA-1-81 (FIGS. 2A-C, 5A-C and 6A-D). Additionally, the level of transcripts of specific undifferentiation markers (e.g., Oct 4, Nanog, Sox2, Rex1, Cx43, FGF4) or differentiation markers (e.g., albumin, glucagons, α -cardiac actin, β -globulin, Flk1, AC133 and neurofilament) can be detected using RNA-based techniques such as RT-PCR analysis and/or cDNA microarray analysis.

(174) Determination of ES cell differentiation can also be effected via measurements of alkaline phosphatase activity. Undifferentiated human ES cells have alkaline phosphatase activity which can be detected by fixing the cells with 4% paraformaldehyde and developing with the Vector Red substrate kit according to manufacturer's instructions (Vector Laboratories, Burlingame, California, USA).

(175) The present inventors have uncovered that the novel xeno-free and serum free culture media of the invention can be used to derive new pluripotent stem cell lines.

(176) Thus, as is further shown in the Examples section which follows, using the HA40/4 medium culture medium the present inventors were capable of deriving a new hESC line referred to as "WC1" from whole blastocysts cultured on human foreskin fibroblasts feeder layer (Example 3 of the Examples section which follows).

(177) The term "deriving" as used herein refers to generating an embryonic stem cell line or an induced pluripotent stem cell line from at least one embryonic stem or induced pluripotent cell.

(178) According to some embodiments of the invention, the pluripotent stem cell line is an embryonic stem cell line, and the method of deriving the embryonic stem cell line is effected by: (a) obtaining an embryonic stem cell from a pre-implantation stage blastocyst, post-implantation stage blastocyst and/or a genital tissue of a fetus; and (b) culturing the embryonic stem cell in the culture medium of some embodiments of the invention, thereby deriving the embryonic stem cell line.

(179) As used herein the phrase "embryonic stem cell line" refers to embryonic stem cells which are derived from a single or a group of embryonic stem cells of a single organism (e.g., a single human blastocyst), and which are characterized by the ability to proliferate in culture while maintaining the undifferentiated state and the pluripotent capacity.

(180) Obtaining an embryonic stem cell from a pre-implantation stage blastocyst, post-implantation stage blastocyst and/or a genital tissue of a fetus can be performed using methods known in the art, as described hereinabove and in Example 3 of the Examples section which follows. Briefly, the

zona pellucida is removed from a 5-7 day-old blastocyst using Tyrode's acidic solution (Sigma, St Louis MO, USA), the trophoblast layer is specifically removed either by immunosurgery or mechanically using 27 g needles and the exposed ICM is either directly cultured in a suitable culture system (e.g., feeder layers, feeder-free matrix or a suspension culture) in the presence of any of the culture media described hereinabove for 4-10 days (in case a preimplantation blastocyst is used) or subject to in vitro implantation by culturing the ICM for 6-8 days (to obtain cells of a 13 day-old blastocyst in case a post-implantation/pre-gastrulation blastocyst is used) on feeder layers or a feeder-free culturing system which allow implantation of the blastocyst to the surface, following which the implanted cells are isolated and can be further cultured on feeder layers, feeder-free matrix or a suspension culture in the presence of any of the culture media described hereinabove as described hereinunder. When using the genital tissue of a fetus, the genital ridges are dissociated and cut into small chunks which are thereafter disaggregated into cells by mechanical dissociation. The single cell EG cells are then cultured in any of the culture media described hereinabove for 4-10 days.

(181) According to some embodiments of the invention, the pluripotent stem cell line is an induced pluripotent stem cell (iPS cell) line, and the method of deriving the iPS cell line is effected by: (a) inducing a somatic cell to a pluripotent stem cell; and (b) culturing the pluripotent stem cell in the culture medium of some embodiments of the invention, thereby deriving the induced pluripotent stem cell line.

(182) As used herein the phrase "induced pluripotent stem cell line" refers to pluripotent stem cells derived from a single induced pluripotent stem cell, which are characterized by the ability to proliferate in culture while maintaining the undifferentiated state and the pluripotent capacity.

(183) Methods of inducing pluripotent stem cells are well known in the art and examples are given in Takahashi and Yamanaka, 2006; Takahashi et al, 2007; Meissner et al, 2007; Okita et al, 2007, Yu et al, 2007; Nakagawa et al, 2008, Yu J. et al., Science. 2009, 324: 797-801; Park et al, 2008; Hanna et al, 2007; Lowry et al, 2008; Aoi et al, 2008; all of which are fully incorporated by reference herein.

(184) Once obtained the ESCs of iPS cells are further cultured in any of the culture media described hereinabove which allow expansion of the pluripotent stem cells in the undifferentiated state, essentially as described hereinabove.

(185) It will be appreciated that an established pluripotent stem cell line (e.g., embryonic stem cell line or induced pluripotent stem cell line) can be subject to freeze/thaw cycles without hampering the proliferative capacity of the cells in the undifferentiated state while preserving their pluripotent capacity. For example, as is shown in the Examples section which follows, using 15% serum replacement and 10% DMSO, hESCs or human iPS cells were successfully frozen and thawed.

(186) As described in Examples 1, 2, 4 and 5 of the Examples section which follows, hESCs and human iPS cells which were expanded and maintained in any of the culture media described hereinabove are pluripotent (i.e., capable of differentiating into all cell types of the three embryonic germ layers, the ectoderm, the endoderm and the mesoderm) as evidenced in vitro (by the formation of EBs) and in vivo (by the formation of teratomas) after a prolonged culture period (e.g., of at least 20 or 30 passages) in the two-dimensional (e.g., feeder-free matrices or foreskin feeders) or three-dimensional (e.g., static or dynamic suspension cultures) culture systems. Thus, hESCs or human iPS cells cultured according to the teachings of the present invention can be used as a source for generating differentiated, lineage-specific cells. Such cells can be obtained directly from the ESCs by subjecting the ESCs to various differentiation signals (e.g., cytokines, hormones, growth factors) or indirectly, via the formation of embryoid bodies and the subsequent differentiation of cells of the EBs to lineage-specific cells.

(187) Thus, according to an aspect of the some embodiments of the invention there is provided a method of generating embryoid bodies from pluripotent stem cells. The method is effected by (a) culturing the pluripotent stem cells according to the method of some embodiment of the invention

to thereby obtain expanded, undifferentiated pluripotent stem cells; and (b) subjecting the expanded, undifferentiated pluripotent stem cells to culturing conditions suitable for differentiating the stem cells to embryoid bodies, thereby generating the embryoid bodies from the pluripotent stem cells.

(188) As used herein the phrase “embryoid bodies” refers to morphological structures comprised of a population of ESCs, extended blastocyst cells (EBCs), embryonic germ cells (EGCs) and/or induced pluripotent stem cells which have undergone differentiation. EBs formation initiates following the removal of differentiation blocking factors from the pluripotent stem cell cultures. In the first step of EBs formation, the pluripotent stem cells proliferate into small masses of cells which then proceed with differentiation. In the first phase of differentiation, following 1-4 days in culture for either human ESCs or human iPS cells, a layer of endodermal cells is formed on the outer layer of the small mass, resulting in “simple EBs”. In the second phase, following 3-20 days post-differentiation, “complex EBs” are formed. Complex EBs are characterized by extensive differentiation of ectodermal and mesodermal cells and derivative tissues.

(189) Thus, the method according to some embodiments of the invention involves the culturing of the pluripotent stem cells in any of the culture media described hereinabove in order to obtain expanded, undifferentiated pluripotent stem cells and then subjecting the expanded, undifferentiated pluripotent stem cells (e.g., ESCs or iPS cells) to culturing conditions suitable for differentiating the pluripotent stem cells to embryoid bodies. Such culturing conditions are substantially devoid of differentiation inhibitory factors which are employed when pluripotent stem cells are to be expanded in an undifferentiated state, such as TGF β .sub.3, ascorbic acid at a concentration of at least 50 μ g/ml, bFGF and/or the IL6RIL6 chimera.

(190) For EBs formation, the pluripotent stem cells (ESCs or iPS cells) are removed from their feeder cell layers, feeder-free-culturing systems or suspension cultures and are transferred to a suspension culture in the presence of a culture medium containing serum or serum replacement and being devoid of differentiation-inhibitory factors (see e.g., Examples 1, 2, 4 and 5 of the Examples section which follows). For example, a culture medium suitable for EBs formation may include a basic culture medium (e.g., Ko-DMEM or DMEM/F12) supplemented with 20% FBSd (HyClone, Utah, USA), 1 mM L-glutamine, 0.1 mM β -mercaptoethanol, and 1% non-essential amino acid stock.

(191) Monitoring the formation of EBs is within the capabilities of those skilled in the art and can be effected by morphological evaluations (e.g., histological staining) and determination of expression of differentiation-specific markers [e.g., using immunological techniques or RNA-based analysis (e.g., RT-PCR, cDNA microarray)].

(192) It will be appreciated that in order to obtain lineage-specific cells from the EBs, cells of the EBs can be further subjected to culturing conditions suitable for lineage-specific cells.

(193) Preferably, the method of this aspect of the present invention further includes step (c) of subjecting cells of the embryoid bodies to culturing conditions suitable for differentiating and/or expanding lineage specific cells; thereby generating the lineage-specific cells from the embryonic stem cells.

(194) As used herein the phrase “culturing conditions suitable for differentiating and/or expanding lineage specific cells” refers to a combination of culture system, e.g., feeder cell layers, feeder-free matrix or a suspension culture and a culture medium which are suitable for the differentiation and/or expansion of specific cell lineages derived from cells of the EBs. Non-limiting examples of such culturing conditions are further described hereinunder.

(195) According to some embodiments of the invention, the method of this aspect of the invention further includes isolating lineage specific cells following step (b).

(196) As used herein, the phrase “isolating lineage specific cells” refers to the enrichment of a mixed population of cells in a culture with cells predominantly displaying at least one characteristic associated with a specific lineage phenotype. It will be appreciated that all cell lineages are derived

from the three embryonic germ layers. Thus, for example, hepatocytes and pancreatic cells are derived from the embryonic endoderm, osseous, cartilaginous, elastic, fibrous connective tissues, myocytes, myocardial cells, bone marrow cells, vascular cells (namely endothelial and smooth muscle cells), and hematopoietic cells are differentiated from embryonic mesoderm and neural, retina and epidermal cells are derived from the embryonic ectoderm.

(197) According to some preferred embodiments of the invention, isolating lineage specific cells is effected by sorting of cells of the EBs via fluorescence activated cell sorter (FACS).

(198) Methods of isolating EB-derived-differentiated cells via FACS analysis are known in the art. According to one method, EBs are disaggregated using a solution of Trypsin and EDTA (0.025% and 0.01%, respectively), washed with 5% fetal bovine serum (FBS) in phosphate buffered saline (PBS) and incubated for 30 min on ice with fluorescently-labeled antibodies directed against cell surface antigens characteristics to a specific cell lineage. For example, endothelial cells are isolated by attaching an antibody directed against the platelet endothelial cell adhesion molecule-1 (PECAM1) such as the fluorescently-labeled PECAM1 antibodies (30884X) available from PharMingen (PharMingen, Becton Dickinson Bio Sciences, San Jose, CA, USA) as described in Levenberg, S. et al., (Endothelial cells derived from human embryonic stem cells. *Proc. Natl. Acad. Sci. USA*. 2002. 99: 4391-4396). Hematopoietic cells are isolated using fluorescently-labeled antibodies such as CD34-FITC, CD45-PE, CD31-PE, CD38-PE, CD90-FITC, CD117-PE, CD15-FITC, class I-FITC, all of which IgG1 are available from PharMingen, CD133/1-PE (IgG1) (available from Miltenyi Biotec, Auburn, CA), and glycophorin A-PE (IgG1), available from Immunotech (Miami, FL). Live cells (i.e., without fixation) are analyzed on a FACScan (Becton Dickinson Bio Sciences) by using propidium iodide to exclude dead cells with either the PC-LYSIS or the CELLQUEST software. It will be appreciated that isolated cells can be further enriched using magnetically-labeled second antibodies and magnetic separation columns (MACS, Miltenyi) as described by Kaufman, D. S. et al., (Hematopoietic colony-forming cells derived from human embryonic stem cells. *Proc. Natl. Acad. Sci. USA*. 2001, 98: 10716-10721).

(199) According to some embodiments of the invention, isolating lineage specific cells is effected by a mechanical separation of cells, tissues and/or tissue-like structures contained within the EBs.

(200) For example, beating cardiomyocytes can be isolated from EBs as disclosed in U.S. Pat. Appl. No. 20030022367 to Xu et al. Four-day-old EBs of the present invention are transferred to gelatin-coated plates or chamber slides and are allowed to attach and differentiate. Spontaneously contracting cells, which are observed from day 8 of differentiation, are mechanically separated and collected into a 15-mL tube containing low-calcium medium or PBS. Cells are dissociated using Collagenase B digestion for 60-120 minutes at 37° C., depending on the Collagenase activity. Dissociated cells are then resuspended in a differentiation KB medium (85 mM KCl, 30 mM K.sub.2HPO.sub.4, 5 mM MgSO.sub.4, 1 mM EGTA, 5 mM creatine, 20 mM glucose, 2 mM Na.sub.2ATP, 5 mM pyruvate, and 20 mM taurine, buffered to pH 7.2, Maltsev et al., *Circ. Res.* 75:233, 1994) and incubated at 37° C. for 15-30 min. Following dissociation cells are seeded into chamber slides and cultured in the differentiation medium to generate single cardiomyocytes capable of beating.

(201) According to some embodiments of the invention, isolating lineage specific cells is effected by subjecting the EBs to differentiation factors to thereby induce differentiation of the EBs into lineage specific differentiated cells.

(202) Following is a non-limiting description of a number of procedures and approaches for inducing differentiation of EBs to lineage specific cells.

(203) To differentiate the EBs of some embodiments of the invention into neural precursors, four-day-old EBs are cultured for 5-12 days in tissue culture dishes including DMEM/F-12 medium with 5 mg/ml insulin, 50 mg/ml transferrin, 30 nM selenium chloride, and 5 mg/ml fibronectin (ITSFn medium, Okabe, S. et al., 1996, *Mech. Dev.* 59: 89-102). The resultant neural precursors can be further transplanted to generate neural cells in vivo (Brüstle, O. et al., 1997. *In vitro-*

generated neural precursors participate in mammalian brain development. Proc. Natl. Acad. Sci. USA. 94: 14809-14814). It will be appreciated that prior to their transplantation, the neural precursors are trypsinized and triturated to single-cell suspensions in the presence of 0.1% DNase. (204) EBs of some embodiments of the invention can differentiate to oligodendrocytes and myelinate cells by culturing the cells in modified SATO medium, i.e., DMEM with bovine serum albumin (BSA), pyruvate, progesterone, putrescine, thyroxine, triiodothyronine, insulin, transferrin, sodium selenite, amino acids, neurotrophin 3, ciliary neurotrophic factor and Hepes (Bottenstein, J. E. & Sato, G. H., 1979, Proc. Natl. Acad. Sci. USA 76, 514-517; Raff, M. C., Miller, R. H., & Noble, M., 1983, Nature 303: 390-396]. Briefly, EBs are dissociated using 0.25% Trypsin/EDTA (5 min at 37° C.) and triturated to single cell suspensions. Suspended cells are plated in flasks containing SATO medium supplemented with 5% equine serum and 5% fetal calf serum (FCS). Following 4 days in culture, the flasks are gently shaken to suspend loosely adhering cells (primarily oligodendrocytes), while astrocytes are remained adhering to the flasks and further producing conditioned medium. Primary oligodendrocytes are transferred to new flasks containing SATO medium for additional two days. Following a total of 6 days in culture, oligospheres are either partially dissociated and resuspended in SATO medium for cell transplantation, or completely dissociated and a plated in an oligosphere-conditioned medium which is derived from the previous shaking step [Liu, S. et al., (2000). Embryonic stem cells differentiate into oligodendrocytes and myelinate in culture and after spinal cord transplantation. Proc. Natl. Acad. Sci. USA. 97: 6126-6131].

(205) For mast cell differentiation, two-week-old EBs of some embodiments of the invention are transferred to tissue culture dishes including DMEM medium supplemented with 10% FCS, 2 mM L-glutamine, 100 units/ml penicillin, 100 mg/ml streptomycin, 20% (v/v) WEHI-3 cell-conditioned medium and 50 ng/ml recombinant rat stem cell factor (rrSCF, Tsai, M. et al., 2000. In vivo immunological function of mast cells derived from embryonic stem cells: An approach for the rapid analysis of even embryonic lethal mutations in adult mice in vivo. Proc Natl Acad Sci USA. 97: 9186-9190). Cultures are expanded weekly by transferring the cells to new flasks and replacing half of the culture medium.

(206) To generate hemato-lymphoid cells from the EBs of some embodiments of the invention. 2-3 days-old EBs are transferred to gas-permeable culture dishes in the presence of 7.5% CO₂ and 5% O₂ using an incubator with adjustable oxygen content. Following 15 days of differentiation, cells are harvested and dissociated by gentle digestion with Collagenase (0.1 unit/mg) and Dispase (0.8 unit/mg), both are available from F. Hoffman-La Roche Ltd, Basel, Switzerland. CD45-positive cells are isolated using anti-CD45 monoclonal antibody (mAb) M1/9.3.4.HL.2 and paramagnetic microbeads (Miltenyi) conjugated to goat anti-rat immunoglobulin as described in Potocnik, A. J. et al., (Immunology Hemato-lymphoid in vivo reconstitution potential of subpopulations derived from in vitro differentiated embryonic stem cells. Proc. Natl. Acad. Sci. USA. 1997, 94: 10295-10300). The isolated CD45-positive cells can be further enriched using a single passage over a MACS column (Miltenyi).

(207) It will be appreciated that the culturing conditions suitable for the differentiation and expansion of the isolated lineage specific cells include various tissue culture media, growth factors, antibiotic, amino acids and the like and it is within the capability of one skilled in the art to determine which conditions should be applied in order to expand and differentiate particular cell types and/or cell lineages.

(208) Additionally or alternatively, lineage specific cells can be obtained by directly inducing the expanded, undifferentiated pluripotent stem cells such as ESCs or iPS cells to culturing conditions suitable for the differentiation of specific cell lineage.

(209) According to an aspect of some embodiments of the invention there is provided a method of generating lineage-specific cells from pluripotent stem cells. The method is effected by (a) culturing the pluripotent stem cells according to the method of some embodiments of the invention,

to thereby obtain expanded, undifferentiated stem cells; and (b) subjecting the expanded, undifferentiated stem cells to culturing conditions suitable for differentiating and/or expanding lineage specific cells, thereby generating the lineage-specific cells from the pluripotent stem cells. (210) Following are non-limiting examples of culturing conditions which are suitable for differentiating and/or expanding lineage specific cells from pluripotent stem cells (e.g., ESCs and iPS cells).

(211) Mesenchymal stromal cells which are CD73-positive and SSEA-4-negative can be generated from hESCs by mechanically increasing the fraction of fibroblast-like differentiated cells formed in cultures of hESCs, essentially as described in Trivedi P and Hematti P. *Exp Hematol*. 2008, 36(3):350-9. Briefly, to induce differentiation of hESC the intervals between medium changes are increased to 3-5 days, and the cells at the periphery of the ESC colonies become spindle-shaped fibroblast-looking cells. After 9-10 days under these conditions when about 40-50% of the cells in the culture acquire the fibroblast-looking appearance, the undifferentiated portions of ESC colonies are physically removed and the remaining differentiated cells are passaged to new culture plates under the same conditions.

(212) To induce differentiation of hESCs into dopaminergic (DA) neurons, the cells can be co-cultured with the mouse stromal cell lines PA6 or MS5, or can be cultured with a combination of stromal cell-derived factor 1 (SDF-1/CXCL12), pleiotrophin (PTN), insulin-like growth factor 2 (IGF2) and ephrin B1 (EFNB1) essentially as described in Vazin T, et al., *PLoS One*. 2009 Aug. 12; 4(8):e6606; and in Elkabetz Y., et al., *Genes Dev*. 2008 Jan. 15; 22: 152-165.

(213) To generate mesencephalic dopamine (mesDA) neurons, hESCs can be genetically modified to express the transcription factor Lmx1a (e.g., using a lentiviral vector with the PGK promoter and Lmx1a) essentially as described in Friling S., et al., *Proc Natl Acad Sci USA*. 2009, 106: 7613-7618.

(214) To generate lung epithelium (type II pneumocytes) from hESCs, the ESCs can be cultured in the presence of a commercially available cell culture medium (Small Airway Growth Medium; Cambrex, College Park, MD), or alternatively, in the presence of a conditioned medium collected from a pneumocyte cell line (e.g., the A549 human lung adenocarcinoma cell line) as described in Rippon H J., et al., *Proc Am Thorac Soc*. 2008;5:717-722.

(215) To induce differentiation of hESCs or human iPS cells into neural cells, the pluripotent stem cells can be cultured for about 5 days in the presence of a serum replacement medium supplemented with TGF- β inhibitor (SB431542, Tocris; e.g., 10 nM) and Noggin (R&D; e.g., 500 ng/ml), following which the cells are cultured with increasing amounts (e.g., 25%, 50%, 75%, changed every two days) of N2 medium (Li X J., et al., *Nat Biotechnol*. 2005, 23:215-21) in the presence of 500 ng/mL Noggin, essentially as described in Chambers S M., et al., *Nat Biotechnol*. 2009, 27: 275-280.

(216) In addition to the lineage-specific primary cultures, EBs of the invention can be used to generate lineage-specific cell lines which are capable of unlimited expansion in culture.

(217) Cell lines of the present invention can be produced by immortalizing the EB-derived cells by methods known in the art, including, for example, expressing a telomerase gene in the cells (Wei, W. et al., 2003. *Mol Cell Biol*. 23:2859-2870) or co-culturing the cells with NIH 3T3 hph-HOX11 retroviral producer cells (Hawley, R. G. et al., 1994. *Oncogene* 9: 1-12).

(218) It will be appreciated that since the lineage-specific cells or cell lines obtained according to the teachings of the invention are developed by differentiation processes similar to those naturally occurring in the human embryo they can be further used for human cell-based therapy and tissue regeneration.

(219) Thus, the invention envisages the use of the expanded and/or differentiated lineage-specific cells or cell lines of some embodiments of the invention for treating a disorder requiring cell replacement therapy.

(220) For example, oligodendrocyte precursors can be used to treat myelin disorders (Repair of

myelin disease: Strategies and progress in animal models. *Molecular Medicine Today*. 1997. pp. 554-561), chondrocytes or mesenchymal cells can be used in treatment of bone and cartilage defects (U.S. Pat. No. 4,642,120) and cells of the epithelial lineage can be used in skin regeneration of a wound or burn (U.S. Pat. No. 5,716,411).

(221) For certain disorders, such as genetic disorders in which a specific gene product is missing [e.g., lack of the CFTR gene-product in cystic fibrosis patients (Davies J C, 2002. New therapeutic approaches for cystic fibrosis lung disease. *J. R. Soc. Med.* 95 Suppl 41:58-67)], ESC-derived cells or iPS cells-derived cells are preferably manipulated to over-express the mutated gene prior to their administration to the individual. It will be appreciated that for other disorders, the ESC-derived cells or iPS-derived cells should be manipulated to exclude certain genes.

(222) Over-expression or exclusion of genes can be effected using knock-in and/or knock-out constructs [see for example, Fukushima, S. and Ikeda, J. E.: Trapping of mammalian promoters by Cre-lox site-specific recombination. *DNA Res* 3 (1996) 73-50; Bedell, M. A., Jerkins, N. A. and Copeland, N. G.: Mouse models of human disease. Part I: Techniques and resources for genetic analysis in mice. *Genes and Development* 11 (1997) 1-11; Bermingham, J. J., Scherer, S. S., O'Connell, S., Arroyo, E., Kalla, K. A., Powell, F. L. and Rosenfeld, M. G.: Tst-1/Oct-6/SCIP regulates a unique step in peripheral myelination and is required for normal respiration. *Genes Dev* 10 (1996) 1751-62].

(223) In addition to cell replacement therapy, the lineage specific cells of some embodiments of the invention can also be utilized to prepare a cDNA library. mRNA is prepared by standard techniques from the lineage specific cells and is further reverse transcribed to form cDNA. The cDNA preparation can be subtracted with nucleotides from embryonic fibroblasts and other cells of undesired specificity, to produce a subtracted cDNA library by techniques known in the art.

(224) The lineage specific cells of some embodiments of the invention can be used to screen for factors (such as small molecule drugs, peptides, polynucleotides, and the like) or conditions (such as culture conditions or manipulation) that affect the differentiation of lineage precursor to terminally differentiated cells. For example, growth affecting substances, toxins or potential differentiation factors can be tested by their addition to the culture medium.

(225) As used herein the term "about" refers to $\pm 10\%$.

(226) The terms "comprises", "comprising", "includes", "including". "having" and their conjugates mean "including but not limited to".

(227) The term "consisting of" means "including and limited to".

(228) The term "consisting essentially of" means that the composition, method or structure may include additional ingredients, steps and/or parts, but only if the additional ingredients, steps and/or parts do not materially alter the basic and novel characteristics of the claimed composition, method or structure.

(229) As used herein, the singular form "a", "an" and "the" include plural references unless the context clearly dictates otherwise. For example, the term "a compound" or "at least one compound" may include a plurality of compounds, including mixtures thereof.

(230) Throughout this application, various embodiments of this invention may be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the invention. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 3, 4, 5, and 6. This applies regardless of the breadth of the range.

(231) Whenever a numerical range is indicated herein, it is meant to include any cited numeral (fractional or integral) within the indicated range. The phrases "ranging/ranges between" a first

indicate number and a second indicate number and “ranging/ranges from” a first indicate number “to” a second indicate number are used herein interchangeably and are meant to include the first and second indicated numbers and all the fractional and integral numerals therebetween.

(232) As used herein the term “method” refers to manners, means, techniques and procedures for accomplishing a given task including, but not limited to, those manners, means, techniques and procedures either known to, or readily developed from known manners, means, techniques and procedures by practitioners of the chemical, pharmacological, biological, biochemical and medical arts.

(233) As used herein, the term “treating” includes abrogating, substantially inhibiting, slowing or reversing the progression of a condition, substantially ameliorating clinical or aesthetical symptoms of a condition or substantially preventing the appearance of clinical or aesthetical symptoms of a condition.

(234) It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable subcombination or as suitable in any other described embodiment of the invention. Certain features described in the context of various embodiments are not to be considered essential features of those embodiments, unless the embodiment is inoperative without those elements.

(235) Various embodiments and aspects of the present invention as delineated hereinabove and as claimed in the claims section below find experimental support in the following examples.

EXAMPLES

(236) Reference is now made to the following examples, which together with the above descriptions illustrate some embodiments of the invention in a non limiting fashion.

(237) Generally, the nomenclature used herein and the laboratory procedures utilized in the present invention include molecular, biochemical, microbiological and recombinant DNA techniques. Such techniques are thoroughly explained in the literature. See, for example, “Molecular Cloning: A laboratory Manual” Sambrook et al., (1989); “Current Protocols in Molecular Biology” Volumes I-III Ausubel, R. M., ed. (1994); Ausubel et al., “Current Protocols in Molecular Biology”, John Wiley and Sons, Baltimore, Maryland (1989); Perbal, “A Practical Guide to Molecular Cloning”, John Wiley & Sons, New York (1988); Watson et al., “Recombinant DNA”, Scientific American Books, New York; Birren et al. (eds) “Genome Analysis: A Laboratory Manual Series”, Vols. 1-4, Cold Spring Harbor Laboratory Press, New York (1998); methodologies as set forth in U.S. Pat. Nos. 4,666,828; 4,683,202; 4,801,531; 5,192,659 and 5,272,057; “Cell Biology: A Laboratory Handbook”, Volumes I-III Cellis, J. E., ed. (1994); “Current Protocols in Immunology” Volumes I-III Coligan J. E., ed. (1994); Stites et al. (eds), “Basic and Clinical Immunology” (8th Edition), Appleton & Lange, Norwalk, CT (1994); Mishell and Shiigi (eds), “Selected Methods in Cellular Immunology”, W. H. Freeman and Co., New York (1980); available immunoassays are extensively described in the patent and scientific literature, see, for example, U.S. Pat. Nos. 3,791,932; 3,839,153; 3,850,752; 3,850,578; 3,853,987; 3,867,517; 3,879,262; 3,901,654; 3,935,074; 3,984,533; 3,996,345; 4,034,074; 4,098,876; 4,879,219; 5,011,771 and 5,281,521; “Oligonucleotide Synthesis” Gait, M. J., ed. (1984); “Nucleic Acid Hybridization” Hames, B. D., and Higgins S. J., eds. (1985); “Transcription and Translation” Hames, B. D., and Higgins S. J., Eds. (1984); “Animal Cell Culture” Freshney, R. I., ed. (1986); “Immobilized Cells and Enzymes” IRL Press, (1986); “A Practical Guide to Molecular Cloning” Perbal, B., (1984) and “Methods in Enzymology” Vol. 1-317, Academic Press; “PCR Protocols: A Guide To Methods And Applications”, Academic Press, San Diego, CA (1990); Marshak et al., “Strategies for Protein Purification and Characterization—A Laboratory Course Manual” CSHL Press (1996); all of which are incorporated by reference as if fully set forth herein. Other general references are provided throughout this document. The procedures therein are believed to be well known in the art and are

provided for the convenience of the reader.

(238) All the information contained therein is incorporated herein by reference.

General Materials and Experimental Methods

(239) Cell Lines

(240) iPS cell culture—Induced pluripotent stem (iPS) cell lines J1.2-3 and iF4 [Park et al, 2008] derived from foreskin fibroblast and adult fibroblasts respectively, were cultured with inactivated mouse embryonic fibroblasts (MEF) as was previously described [Park et al, 2008]. The following culture media combinations were tested for their ability to support the growth of iPS cells in attached [two-dimensional (2D)] cultures:

(241) hESC culture—The human ESC lines I4, I3, I6 and H9.2 were used in the study.

(242) Culturing conditions on two-dimensions: hESCs lines or human iPS cell lines were cultured with MEFs or on synthetic matrices in the presence of the tested culture media. Cells were passaged every four to six days using 1 mg/ml type IV collagenase (Gibco Invitrogen Corporation, Grand Island NY, USA) and plated at a density of 1×10^4 – 3×10^5 cells per cm^2 .

(243) Media Used for 2D Cultures—

(244) (i) yF10 basic culture medium consisting of 85% DMEM/F12 (Biological Industries, Beit Haemek, Israel), 15% KNOCKOUT™ serum replacement (SR; Invitrogen), 2 mM L-glutamine, 0.1 mM β -mercaptoethanol, 1% non-essential amino acid stock, and 10 ng/ml basic fibroblast growth factor (bFGF), all from Invitrogen Corporation products, Grand Island NY, USA, unless otherwise indicated. This basic culture medium was used as a control and for the routine growth of iPS cells or hESCs with inactivated MEF or foreskin fibroblasts as feeder layers in 2D cultures.

(245) (ii) mHA40/4 DMEM/F12 (94%) (Biological Industries, Israel), ITS 1% [Invitrogen corporation; the ITS premix is a $\times 100$ stock solution consists of 1.25 mg Insulin, 1.25 mg Transferrin and 1.25 mg Selenius acid], 2 ng/ml TGF β .sub.3 (from R&D Systems Minneapolis MN, USA), L-glutamine 2 mM (Invitrogen corporation), ascorbic acid 500 $\mu\text{g/ml}$ (Sigma, Israel), bFGF—10 ng (Invitrogen corporation), Human serum albumin—0.5% (Sigma, Catalogue No. A1653). Na-Bicarbonate (7.5%) (Biological Industries, Israel), defined lipid mixture 1% (Invitrogen corporation).

(246) (iii) HA75 DMEM/F12 (94%) (Biological Industries, Israel), L-glutamine 2 mM (Invitrogen corporation), ascorbic acid 500 $\mu\text{g/ml}$ (Sigma), bFGF—10 ng (Invitrogen Corporation), TGF β .sub.3 2 ng/ml (R&D Systems Minneapolis MN, USA). SR3 (serum replacement)—1% (Sigma, Israel), defined lipid mixture 1% (Invitrogen corporation).

(247) (iv) HA76 DMEM/F12 (94%) (Biological Industries. Beit HaEmek, Israel). ITS 1% (Invitrogen corporation), L-glutamine 2 mM (Invitrogen corporation), ascorbic acid 500 $\mu\text{g/ml}$ (Sigma, Israel), bFGF—100 ng (Invitrogen corporation), TGF β .sub.3 2 ng/ml (R&D Systems Minneapolis MN, USA), Human serum albumin serum—1% (Sigma, Catalogue No. A1653). Na-Bicarbonate (7.5%) (Biological Industries, Israel), defined lipid mixture 1% (Invitrogen corporation).

(248) (v) HA77 DMEM/F12 (94%) (Biological Industries. Israel, Sigma Israel), L-glutamine 2 mM (Invitrogen corporation, Sigma, Israel), ascorbic acid 500 $\mu\text{g/ml}$ (Sigma, Israel), bFGF—100 ng (Invitrogen corporation), Na-Bicarbonate (7.5%) (Biological Industries, Israel). SR3—1% (Sigma, Israel), defined lipid mixture 1% (Invitrogen corporation, Sigma, Israel). It should be noted that the HA77 DMEM/F12 (94%) can also be used without Na-Bicarbonate at all and yet support the culture of pluripotent stem cells (e.g., hESCs and iPSCs) in a proliferative, pluripotent and undifferentiated state for at least 10 passages.

(249) (vi) HA78 DMEM/F12 (94%) (Biological Industries, Israel), L-glutamine 2 mM (Invitrogen corporation), ascorbic acid 500 $\mu\text{g/ml}$ (Sigma, Israel), bFGF—10 ng/ml (Invitrogen corporation), TGF β .sub.3 2 ng/ml (R&D Systems Minneapolis MN, USA), SR3™—1% (Sigma, Israel), Na-Bicarbonate (7.5%) (Biological Industries, Israel), defined lipid mixture 1% (Invitrogen corporation).

(250) (v) HA74/1 DMEM/F12 (94%) (Biological Industries, Israel), ITS 1% (Invitrogen corporation), L-glutamine 2 mM (Invitrogen corporation), ascorbic acid 500 µg/ml (Sigma, Israel), bFGF—50 ng/ml (Invitrogen corporation), TGFβ.sub.3 2 ng/ml (R&D Systems Minneapolis MN, USA), Human serum albumin—0.5% (Sigma, Israel, Catalogue No. A1653), Na-Bicarbonate (7.5%) (Biological Industries, Israel), defined lipid mixture 1% (Invitrogen Corporation).

(251) It should be noted that when recombinant human Albumin (SIGMA, Catalogue No. A7223) was used instead of human serum Albumin (SIGMA, Catalogue No. A1653) in the mHA40/4, HA76, HA74/1 culture media, these culture media were found to support the growth of hESCs and iPS cells in a pluripotent and undifferentiated state for an extended period of culture. Thus, these results demonstrate that recombinant human albumin can be used instead of human serum albumin in the culture media of some embodiments of the invention and thereby provide defined, xeno-free conditions.

(252) Culturing conditions in three-dimension culture systems (suspension culture):

(253) Media used for suspension cultures—

(254) (i) CM100Fp medium consisting of the basic culture medium (yF10 basic culture medium) supplemented with 100 pg/ml IL6RIL6 chimera. The 85-Kda IL6RIL6 was produced and purified as described and was donated by InterPharm, Merck-Serono group (Nes-Ziona, Israel and Geneva, Switzerland).

(255) (ii) CM100F medium consisting of the basic culture medium (yF10 basic culture medium) supplemented with 100 ng/ml IL6RIL6 chimera. The 85-Kda IL6RIL6 was produced and purified as described and was donated by InterPharm, Merck-Serono group (Nes-Ziona, Israel and Geneva, Switzerland).

(256) (iii) yF100 basic medium (yF10 basic culture medium) in which instead of 10 ng/ml bFGF 100 ng/ml bFGF was used. This medium was found to support hESCs suspension culture with the same efficiency as CM100F.

(257) (iv) yFL3 medium consists of the yF10 basic culture medium with 4 ng/ml bFGF instead of 10 ng/ml bFGF, and supplemented with 3000 units/ml leukemia inhibitory factor (LIF). It should be noted that iPS cells were cultured with the yFL3 medium which comprised 4 or 10 ng/ml bFGF with the same efficiency.

(258) (v) modified HA13(a) medium consists of DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 500 µg/ml, bFGF—4 ng, and SR3—1%]. Was found to support hESCs and iPSCs in a 2-dimensional and 3-dimensional culture systems.

(259) (vi) modified HA13(b) medium consists of DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 500 µg/ml, bFGF—4 ng, SR3—1% and a lipid mixture (1%)]. Was found to support hESCs and iPSCs in a 2-dimensional and 3-dimensional culture systems.

(260) (vii) modified HA13(c) medium consists of DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 50 µg/ml, bFGF—4 ng, and SR3—1%. Was found to support hESCs and iPSCs in a 2-dimensional and 3-dimensional culture systems.

(261) (viii) modified HA13(d) medium consists of DMEM/F12 (95%), L-glutamine 2 mM, ascorbic acid 50 µg/ml, bFGF—4 ng, SR3—1% and a lipid mixture (1%)]. Was found to support hESCs and iPSCs in a 2-dimensional and 3-dimensional culture systems.

(262) Culture in static suspension (3-dimensional) cultures—To initiate suspension cultures, the iPS cells or the hESCs were removed from their culture dish using 1.5 mg/ml type IV collagenase (Worthington biochemical corporation, Lakewood, NJ, USA), further broken into small clumps using 200 µl Gilson pipette tips, and cultured in suspension in 58 mm Petri dishes (Greiner, Frickenhausen, Germany) at a cell density of 1×10^6 – 5×10^6 cells/dish (5–8 ml of medium in 58-mm dishes). The Petri dishes were kept static in an incubator at 37° C. in 5% CO₂. When required, differentiating clumps were removed from the culture during the first three passages while the cells adapted to the new culture conditions. The medium in the suspension culture was changed daily, and the cells were passaged every 5–7 days either by manual cutting of

clumps using 27 g needles (only at passages 1-3) or by gentle pipetting using 200 ml Gilson pipette tips. Alternatively, the cells were passaged using trypsin EDTA (0.25%, Biological Industries, Beit Haemek, Israel) combined with a one-hour treatment with 10 M ROCK inhibitor (EMD Biosciences, Inc. La Jolla, CA, USA) before the incubation with trypsin. For calculating cells' doubling time the I3, I4 and H9.2 hESCs and the J1.2-3 and iF4 iPS cells were counted and grown in suspension for 8 days with CM100F or CM100Fp culture media. Cells were counted every other day. Average doubling time of four biological repeats was calculated.

(263) Culture in spinner flasks (3-dimensional)—iPS cells or hESCs clumps were cultured in static Petri dishes for at least one passage, and were then transferred to 250 ml spinner flasks (Cell Spin 250 or 100, Integra BioSciences) in the tested culture media, shaken continuously at 90 rounds per minutes (rpm) using a magnetic plate, and placed in a 37° C. in 5% CO₂ incubator. Medium was changed every other day. Every 5-7 days the clumps were split with a ratio of 1:2.

(264) Immunofluorescence of cells cultured on 2-D or 3-D culture systems—For fluorescent immunostaining undifferentiated hESCs or iPS grown in 2-D or 3-D culture systems in the presence of the tested culture media or re-cultured on MEFs were fixed with 4% paraformaldehyde and exposed to the primary antibodies overnight at 4° C. Cys 3 conjugated antibodies (Chemicon International, Temecula CA, USA) were used as secondary antibodies (1:200). The primary antibodies (1:50) include SSEA 1, 3 and 4 (Hybridoma Bank, Iowa, USA), TRA1-60 and TRA-1-81 (Chemicon International, Temecula CA, USA) and Oct4 (Santa Cruz Biotechnology, Santa Cruz, CA, USA).

(265) Immunohistochemistry of iPS cells or hESCs cultured on 2-D or 3-D culture systems—After deparaffinization the tissue sections were stained using Dako LSAB+ staining kit for presence of markers of ectoderm (β -3-tubulin 1:500, Chemicon International, Temecula CA USA), mesoderm (CD31 1:20), and endoderm (α -fetoprotein 1:20) (both from DakoCytomation, Glostrup, Denmark). As controls, both IgG isotype and secondary antibody staining were performed. The secondary antibody was conjugated to peroxidase.

(266) Karyotype analysis of cells cultured on 2-D or 3-D culture systems—Karyotype analysis (G-banding) was performed on at least 10 cells from each sample, two samples per test, as previously described [Amit et al, 2003]. Karyotypes were analyzed and reported according to the “International System for Human Cytogenetic Nomenclature” (ISCN).

(267) Embryoid bodies (EBs) formation of cells cultured on 2-D or 3-D culture systems—For the formation of EBs, hESCs or iPS were passaged as described and transferred to 58 mm Petri dishes (Greiner, Frickenhausen, Germany). EBs were grown in medium consisting of 80% DMEM/F12 (Biological Industries, Beit Haemek, Israel), supplemented with 10% fetal bovine serum (FBS) (HyClone, Utah, USA), 10% KNOCKOUT™ serum replacement (SR; Invitrogen), 2 mM L-glutamine, 0.1 mM β -mercaptoethanol, and 1% non-essential amino acid stock (Invitrogen Corporation, Grand Island NY, USA). 10-14 day-old EBs were harvested for RNA isolation and histological examination. For histological analysis EBs were fixed in 10% neutral-buffered formalin, dehydrated in graduated alcohol (70%-100%) and embedded in paraffin. □ 1-5 μ m sections were deparaffinized and stained with hematoxylin/eosin (H&E).

(268) RT-PCR of cells cultured on 2-D or 3-D culture systems—Total RNA was isolated from hESCs or iPS grown for 10, 15 and 20 passages on the xeno-free two-dimensional or three-dimensional culture systems in the tested media and from 10-14 day-old EBs (formed from cells grown on 2-D, 3-D in the presence of the tested culture media or cells cultured on MEFs) using Tri-Reagent (Sigma, St. Louis MO, USA), according to the manufacturer's instructions. cDNA was synthesized from 1 μ g total RNA using MMLV reverse transcriptase RNase H minus (Promega, Madison WI, USA). PCR reaction included denaturation for 5 minutes at 94° C. followed by repeated cycles of 94° C. for 30 seconds, annealing temperature as indicated in Table 1, hereinbelow, for 30 seconds and extension at 72° C. for 30 seconds. PCR primers and reaction conditions are described in Table 1, hereinbelow. PCR products were size-fractionated using 2%

agarose gel electrophoresis. DNA markers were used to confirm the size of the resultant fragments. For quantitative PCR (Q-PCR), densitometry of tested genes was normalized to GAPDH. Three repeats were conducted for each tested line.

(269) TABLE-US-00001 TABLE 1 RT-PCR primers and conditions Table 1: RT-PCR primers and PCR conditions are provided along with the GenBank Accession numbers of the amplified transcripts. Gene product Forward (F) (Accession and reverse (R) primers (SEQ ID NO:) Reaction Size number) provided in a 5'.fwdarw.3' direction Condition (bp) Oct-4 F: 5'-GAGAACAATGAGAACCTTCAGGA (SEQ ID 30 cycles 219 (S81255) NO: 1.) at 60° C. R: 5'-TTCTGCGCCGGTTACAGAACCA (SEQ ID in 1.5 mM NO: 2) MgCl.sub.2 Albumin F: 5'-TGCTTGAATCAGCTGATGACAGGG (SEQ ID 35 cycles 302 (AF542069) NO: 3) at 60° C. R: 5'-AAGGCAAGTCAGCAGCCATGTCAT (SEQ ID in 1.5 mM NO: 4) MgCl.sub.2 α -fetoprotein F: 5'-GCTGGATTCTTCTGCAGGATGGGGAA (SEQ ID 30 cycles 216 (BC027881) NO: 5) at 60° C. R: 5'-TCCCCTGAAGAAAATTGGTTAAAAT (SEQ ID in 1.5 mM NO: 6) MgCl.sub.2 NF-68KD F: 5'-GAGTGAAATGGCACGATACCTA (SEQ ID NO: 7) 30 cycles 473 (NFH R: 5'-TTTCCTCTCCTTCTTCACCTTC (SEQ ID NO: 8) at 60° C. (AY156690; in 2 mM X15307; MgCl.sub.2 X15309) α -cardiac F: 5'-GGAGTTATGGTGGGTATGGGTC (SEQ ID NO: 9) 35 cycles 486 actin R: 5'-AGTGGTGACAAAGGAGTAGCCA (SEQ ID at 65° C. (NM_005159) NO: 10) in 2 mM MgCl.sub.2 β -Actin F: 5'-ATCTGGCACCAACCTTCTACAATGAGCTGCG 35 cycles 838 (NM_001101) (SEQ ID NO: 11) at 62° C. R: 5'-CGTCATACTCCTGCTTGCTGATCCACATCTGC in 1.5 mM (SEQ ID NO: 12) MgCl.sub.2 Sox2 F: 5'CCCCCGGCGGCAATAGCA (SEQ ID NO: 13) 35 cycles 448 (Z31560) R: 5'TCGGCGCCGGGAGATACAT (SEQ ID NO: 14) at 60° C. in 1.5 mM MgCl.sub.2 Rex1 F: 5'GCGTACGCAAATTAAGTCCAGA (SEQ ID 35 cycles 306 (AF450454) NO: 15) at 56° C. R: 5'CAGCATCCTAACAGCTCGCAGAAT (SEQ ID in 1.5 mM NO: 16) MgCl.sub.2 CX43 F: 5'TACCATGCGACCAGTGGTGCGCT (SEQ ID 35 cycles 295 (NM_000165) NO: 17) at 61° C. R: 5'GAATTCTGGTTATCATCGGGGAA (SEQ ID in 1.5 mM NO: 18) MgCl.sub.2 FGF4 F: 5'CTACAACGCCTACGAGTCCTACA (SEQ ID 35 cycles 370 (NM_002007) NO: 19) at 52° C. R: 5'GTTGCACCAGAAAAGTCAGAGTTG (SEQ ID in 1.5 mM NO: 20) MgCl.sub.2 Glucagon F: 5'CTCAGTGATCCTGATCAGATGAACG (SEQ ID 35 cycles 370 (X03991) NO: 21) at 65° C. R: 5'AGTCCCTGGCGGCAAGATTATCAAG (SEQ ID in 1.5 mM NO: 22) MgCl.sub.2 β -globulin F: 5'ACCTGACTCCTGAGGAGAAGTCTGC (SEQ ID 35 cycles 410 (V00499) NO: 23) at 65° C. R: 5'TAGCCACACCAGCCACCACTTTCTG (SEQ ID in 1.5 mM NO: 24) MgCl.sub.2 Flk1 F: 5'ATGCACGGCATCTGGGAATC (SEQ ID NO: 25) 35 cycles 537 (NM_002253) R: 5'GCTACTGTCCTGCAAGTTGCTGTC (SEQ ID at 65° C. NO: 26) in 1.5 mM MgCl.sub.2 AC133 F: 5'CAGTCTGACCAGCGTGAAAA (SEQ ID NO: 27) 35 cycles 200 (NM_006017) R: 5'GGCCATCCAAATCTGTCCTA (SEQ ID NO: 28) at 65° C. in 1.5 mM MgCl.sub.2 Nanog F: 5'ACTAACATGAGTGTGGATCC (SEQ ID NO: 29) 35 cycles 800 (NG_004095) R: 5'TCATCTTCACACGTCTTCAG (SEQ ID NO :30) at 61° C. in 1.5 mM MgCl.sub.2

(270) Teratoma formation from cells cultured on 2-D—hESCs (H9.2 and I3) and iPS (iF4 and J1.2-3) cells from 4-6 wells of a 6-well plate (each well has 10 cm total surface area and includes 1.5-2.5 $\times$ 10⁶ cells) were harvested and injected into the hindlimb muscles of four week-old male of severe combined immunodeficiency (SCID)-beige mice. Ten weeks after the injection the resultant teratomas were harvested and prepared for histological analysis using the same method mentioned for EBs.

(271) Teratoma formation from cells cultured in suspension (3-D culture systems)—hESCs (H9.2 and 13) and iPS (iF4 and J1.2-3) cells from four to six 58 mm dishes (from suspension culture, each dish includes $1.5\text{--}2.5 \times 10^6$ cells) were harvested and injected into the hindlimb muscles of four week-old male of severe combined immunodeficiency (SCID)-beige mice. Ten weeks after the injection the resultant teratomas were harvested and prepared for histological analysis using the same method mentioned for EBs.

Example 1

Induced Pluripotent Stem Cells and Embryonic Stem Cells can be Maintained in an Undifferentiated and Pluripotent State when Cultured on Xeno-Free, Feeder-Layer-Free 2-D Culture Systems

(272) The experiments described hereinbelow were performed using iPS cells or hESCs which were cultured according to the methods, culturing conditions and culture media described in the “General Materials and Experimental Methods” section above.

(273) Experimental Results

(274) iPS cells and human ESCs cultured on 2D culture systems using xeno-free, serum-free medium and supportive-layers free system exhibit undifferentiated morphology and characteristics typical to iPS or hESCs—Several possible medium combinations (HA74/1, HA75, HA76, HA77, HA78, HA40/4) were tested for the ability to support feeder-layer free and xeno-free (devoid of any animal contaminant) cultures of iPS cells or hESCs. All tested media (i.e., HA74/1, HA75, HA76, HA77, HA78, HA40/4) were found suitable for supporting iPS or hESC cultures for at least 15 passages. Using the tested media under feeder-layer free conditions using a Matrigel™ synthetic matrix iPS cells or hESCs were cultured continuously for at least 15 passages while maintaining their iPS or hESCs features including undifferentiated proliferation, karyotype stability and pluripotency (data not shown). No morphological differences could be observed between colonies grown in the tested culture systems and those grown on MEF in the presence of the yF10 medium, correspondingly, morphological features remained unchanged on a single-cell level, rendering cells small and round, exhibiting high nucleus-to-cytoplasm ratio, with a notable presence of one to three nucleoli and typical spacing between the cells (data not shown). Similar to cells grown on MEFs in the presence of a control medium (yF10 basic culture medium), iPS cells or hESCs which were cultured on a Matrigel™ (BD Bioscience) synthetic matrix in the presence of all of the tested media (HA74/1, HA75, HA76, HA77, HA78, HA40/4) were passaged routinely every five to seven days, at the same ratio of 1 to 2, 2 to 3, or 1 to 3, indicating a similar population doubling time as iPS or hESCs grown on MEFs with the control medium. The iPS cells or the hESCs were passaged at a same seeding efficiency of about 1 million cells per 10 cm^2 , with the same viability rate of over 90%. Using 15% serum replacement (SR) and 10% DMSO, the iPS cells or the hESCs were successfully frozen and thawed.

(275) iPS cells or hESCs cultured on 2D culture systems in animal free medium and supportive layer free system express markers of pluripotency—Several surface markers typical of primate undifferentiated ESCs and iPS cells were examined using immunofluorescent staining [as described in Thomson et al, 1995, 1996, 1998]. Cells cultured with the tested media for at least 15 passages were found to be strongly positive to surface markers SSEA4, TRA-1-60, TRA-1-81 and Oct 4 (data not shown). As in other primate ESCs, staining with SSEA3 was weak and staining for SSEA1 was negative (data not shown).

(276) iPS cells or hESCs cultured on 2D culture systems in animal free medium and supportive layer free systems form EBs in vitro and teratomas in vivo—The developmental potential of the cells after prolonged culture in the tested conditions was examined in vitro by the formation of embryoid bodies (EBs). iPS or hESCs cells cultured in feeder layer-free culture systems in the presence of the tested culture media (HA74/1, HA75, HA76, HA77, HA78, HA40/4) formed EBs similar to those created by ESCs grown on MEFs (data not shown). For example, the ability of iPS cells to form EBs was shown after 28 passages in the HA40/4 medium and 20 passages in the

HA77 medium. Within these EBs, the iPS cells or the hESCs differentiated into cell types representative of the three embryonic germ layers [Itskovitz-eldor et al, 2000]. Following their injection to SCID Beige mice, the iPS cells or the hESCs cultured under the tested conditions formed teratomas containing cell types representative of the three embryonic germ layers (data not shown), thus demonstrating their full pluripotency. For example, the ability of iPS cells to form teratomas was shown after 31 passages in the mHA40/4 medium; after 24 passages in the HA74/1 medium; and after 16 passages in the HA77 medium.

Example 2

Induced Pluripotent Stem Cells and Embryonic Stem Cells can be Maintained in an Undifferentiated and Pluripotent State When Cultured on Xeno-Free Feeder-Layers in the Presence of Xeno-Free and Serum-Free Medium

(277) The experiments described hereinbelow were performed using iPS cells or hESCs which were cultured according to the methods, culturing conditions and culture media described in the “General Materials and Experimental Methods” section above.

(278) Experimental Results

(279) iPS cells or hESCs cultured on 2D culture systems using xeno-free, serum-free medium and xeno-free feeder cell layers exhibit undifferentiated morphology and characteristics typical to iPS or hESCs—Several possible medium combinations (HA74/1, HA75, HA76, HA77, HA78, HA40\4) were tested for the ability to support xeno-free (devoid of any animal contaminant) cultures of iPS or hESCs using foreskin fibroblast as feeders cell layers. All tested media were found suitable for supporting iPS or hESC cultures. Using the tested media under xeno-free conditions with foreskin fibroblasts as supportive layer, iPS cells or hESCs were cultured continuously for at least 22 passages while maintaining their iPS or hESCs features including undifferentiated proliferation (FIGS. 1A-C and data not shown), karyotype stability and pluripotency. No morphological differences could be observed between colonies grown in the tested culture systems and those grown on MEF in the presence of the control yF10 medium, correspondingly, morphological features remained unchanged on a single-cell level, rendering cells small and round, exhibiting high nucleus-to-cytoplasm ratio, with a notable presence of one to three nucleoli and typical spacing between the cells (data not shown). Similar to cells grown on MEFs, iPS cells or hESCs cultured on foreskin fibroblast feeder cells in the presence of all the tested culture media (HA74/1, HA75, HA76, HA77, HA78, HA40\4) were passaged routinely every five to seven days, at the same ratio of 1 to 2, 2 to 3 or 1 to 3, indicating a similar population doubling time as iPS or hESCs grown on MEFs in the presence of a control yF10 medium. The iPS cells or the hESCs were passaged at a same seeding efficiency of about 1 million cells per 10 cm.^{sup.2}, with the same viability rate of over 90%. Using 15% serum replacement (SR) and 10% DMSO, the iPS cells or the hESCs were successfully frozen and thawed.

(280) iPS cells or hESCs cultured on 2D culture systems in animal free medium and xeno-free supportive layer express markers of pluripotency—Several surface markers typical of primate undifferentiated ESCs and iPS cells were examined using immunofluorescent staining [as described in Thomson et al, 1995, 1996, 1998]. Cells cultured with the tested media for at least 15 passages were found to be strongly positive to surface markers SSEA4, TRA-1-60, TRA-1-81 and Oct 4 (FIGS. 2A-C). As in other primate ESCs, staining with SSEA3 was weak and staining for SSEA1 was negative (data not shown).

(281) iPS cells or hESCs cultured on 2D culture systems in animal free medium and xeno-free feeder layers form EBs in vitro and teratomas in vivo—The developmental potential of the cells after prolonged culture in the tested conditions was examined in vitro by the formation of embryoid bodies (EBs). iPS cells or hESCs cultured in xeno-free feeder cell layers (foreskin fibroblasts) in the presence of the tested culture media (HA74/1, HA75, HA76, HA77, HA78, HA40\4) formed EBs similar to those created by ESCs grown on MEFs in the presence of the yF10 control medium (data not shown). Within these EBs, the iPS cells or hESCs differentiated into cell types

representative of the three embryonic germ layers [Itskovitz-Eldor et al, 2000]. Following their injection to SCID Beige mice, the iPS cells or hESCs cultured under the tested conditions form teratomas containing cell types representative of the three embryonic germ layers (data not shown), thus demonstrating their full pluripotency.

Example 3

Derivation of an Embryonic Stem Cell Line on the Xeno-Free Culture Medium of the Invention
(282) After digestion of the zona pellucida by Tyrode's acidic solution (Sigma, St Louis, MO, USA), whole blastocysts were placed on mitotically inactivated human foreskin fibroblasts (HFF) in the presence of the HA40/4 medium, except that the medium did not contain sodium bicarbonate. Initially, the cells were passage mechanically by using insulin syringes (BD plastipak, Cat. No. 300013) and after 4 passages the cells were passaged every four to six days using 1 mg/ml type IV collagenase (Gibco Invitrogen corporation products, San Diego, CA, USA). The resulting human ESC line was designated "WC1".

Example 4

Induced Pluripotent Stem Cells and Embryonic Stem Cells can be Maintained in an Undifferentiated and Pluripotent State in Static Suspension Cultures

(283) Culture of iPS cells in suspension holds significant advantages over conventional cultures, particularly when aiming to obtain large amounts of cells for cell and tissue transplantation.

(284) The experiments described hereinbelow were performed using iPS cells or hESCs which were cultured according to the methods, culturing conditions and culture media described in the "General Materials and Experimental Methods" section above.

(285) Experimental Results

(286) iPS cells can be maintained in an undifferentiated state in suspension cultures—The iPS cells (the J1.2-3 and iF4 cell lines) which were grown with MEF or in feeder layer-free conditions [Amit et al, 2004], were placed in suspension cultures. After 24 hours in suspension culture with the tested culture medium CM100F, CM100Fp, yFL3 (which comprises 4 ng/ml or 10 ng/ml bFGF and supplemented with 3000 units/ml LIF), or yF100, the iPS cells created spheroid clumps or disc-like structures which were maintained for at least 20 passages (FIGS. 3A-C and data not shown).

Histological examination of the iPS that were cultured in suspension for at least 10 passages revealed a homogenous population of small cells with large nuclei. The spheroids grew and were split mechanically every 5-7 days while maintaining their morphology, allowing expansion of the suspension cultures. Alternatively, by using trypsin-EDTA and ROCK inhibitor treatment, suspended cells could be dissociated into single cells and still formed spheroids of the same morphology and features, thus allowing efficient cell expansion. Some cultures were carried out for over 50 passages (a year of continuous culture). The two different iPS cell lines, J1.2-3 and iF4, which were cultured in suspension as described herein with the tested culture media, showed similar behavior and spheroid morphology and histology.

(287) The yF100 medium (the yF10 basic culture medium which includes 100 ng/ml bFGF instead of 10 ng/ml), the CM100Fp and the yFL3 (the yF10 basic culture medium including 4 ng/ml bFGF instead of 10 ng/ml and supplemented with 3000 units/ml LIF) were found to support the growth of human ESCs in suspension culture in a proliferative, undifferentiated and pluripotent state.

(288) iPS cells which were cultured in suspension and were re-cultured on 2-D culture systems maintain typical iPS cell colony morphology—After at least 10 passages in suspension, when returned to 2D culture with MEFs or fibronectin surface, all of the spheroid clumps adhered to the MEFs or fibronectin surface and after 24-48 hours demonstrated typical iPS cells colony morphology, exhibiting high nucleus-to-cytoplasm ratio with a notable presence of one to three nucleoli and with typical spacing between the cells (FIG. 4).

(289) iPS cells maintain their undifferentiated stem cell phenotype while being cultured in suspension cultures (3D cultures)—Several surface markers typical of primate undifferentiated ESCs and iPS cells were examined using immunofluorescent staining [as described in Thomson et

al, 1998; Bhattacharya, et al. 2004; Kristensen et al 2005]. Human iPS cells which were cultured in suspension with the tested culture media for at least 30 passages were found to be strongly positive for SSEA4, TRA-1-60 and TRA-1-81 and Oct 4 (FIGS. 5A-C). As with other primate ESCs [Thomson et al., 1995 and 1996] and with ESCs cultured with MEFs, staining with SSEA3 was weak and negative for SSEA1 data not shown). Staining for stem cell markers remained high when cells cultured in suspension were returned to 2D cultures with MEFs (data not shown). RT-PCR analyses showed that, similarly to cells cultured with MEFs, iPS cells cultured in suspension for at least 10 passages expressed genetic markers of pluripotency [King et al, 2006] including Oct 4, Nanog, Sox2, Rex1, and FGF4 (data not shown). No significant difference in gene expression of Oct 4, Nanog, Sox2, Rex1, and FGF4 was detected between iPS cells cultured in suspension as compared with iPS cells cultured on MEF, nor with iPS cells that were re-cultured with MEFs after continuous culture in suspension, similar to hESCs under the same conditions.

(290) iPS cells which are cultured in suspension maintain normal karyotype—Karyotype analysis by Giemsa banding was carried out on cells after 30 passages in suspension, and the cells were found to exhibit normal 46,XY karyotype (data not shown). Thus, the karyotype of the suspension cell culture remained stable.

(291) iPS cells or hESCs which are cultured in suspension maintain their pluripotency in vitro—Following prolonged expansion in suspension cultures with the tested culture media, iPS cells or hESCs preserved their pluripotent differentiation ability as was shown by the in vitro formation of EBs. When hESCs or iPS cells which were cultured in suspension for over 20 passages were transferred to serum-containing medium without the addition of the growth factors, formation of cystic EBs was observed after 7-10 days, similarly to cavitated EBs formed from hESCs following 10 days in culture [Itskovitz et al, 2000], and cystic EBs after 14-20 days. Within the EBs formed from the iPS cells or hESCs, there were cell types representative of the three embryonic germ layers typical of iPS cells differentiation (data not shown).

(292) For example, the ability of iPS cells to form EBs was shown after 22 passages in the presence of the CM100p medium in a suspension culture; the ability to form EBs was shown after 23 passages in the presence of the yF100 medium in a suspension culture; the ability to form EBs was shown after 8 passages in the presence of the yFL3 medium in a suspension culture.

(293) iPS cells which are cultured in suspension maintain their pluripotency in vivo—Pluripotency of the suspension iPS cells was further demonstrated in vivo by teratoma formation. Cells cultured in suspension for at least 20 passages were injected into SCID Beige mice, and 10 weeks later tumors were formed (data not shown). Within these teratomas, tissues representative of the three germ layers were observed.

(294) For example, the ability of iPS cells to form teratomas was shown after 20 passages in the CM100 in a suspension culture; and the ability to form teratomas was shown after 10 passages in the yFL3 in a suspension culture.

Example 5

Induced Pluripotent Stem Cells and Embryonic Stem Cells can be Maintained in an Undifferentiated and Pluripotent State in Dynamic Suspension Cultures

(295) The experiments described hereinbelow were performed using iPS cells or hESCs which were cultured according to the methods, culturing conditions and culture media described in the “General Materials and Experimental Methods” section above.

(296) Experimental Results

(297) iPS cells which are cultured in shaking suspension cultures maintain their undifferentiated state—iPS cells from line J1.2-3 or hESCs were cultured in suspension in spinner flasks for at least one month using the tested culture media. An examination after one month showed that the morphological characteristics of the spheroid clumps formed by the cells remained similar to those observed when iPS cells are cultured statically in Petri dishes (data not shown). In addition, the iPS cells strongly expressed markers of undifferentiated hESCs such as Oct-4, TRA-1-81, TRA-1-60

and SSEA4 (FIGS. 6A-D). When re-cultured on MEFs, the iPS cells in the clumps re-attached, forming again typical colonies of iPS cells (data not shown). The karyotype of the cells cultured for one month in the spinner flask was found to be normal (data not shown).

(298) iPS cells which are cultured in dynamic suspension cultures maintain normal karyotype—iPS cells or hESCs which were cultured for 30 passages in static suspension cultures (in the presence of the tested culture media) and then for 3 passages in dynamic (spinner) suspension (in the presence of the tested culture media) were found to exhibit normal 46,XY karyotype. Thus, the karyotype of the suspension iPS cell culture remained stable.

(299) iPS cells or hESCs which are cultured in dynamic suspension maintain their pluripotency in vitro—The developmental potential of the iPS cells or hESCs that were cultured in dynamic suspension cultures was examined in vitro by the formation of EBs, hESCs or iPS were cultured in static suspension for over 20 passages, then on dynamic suspension for at least additional 10 passages, and then were transferred to serum-containing medium without the addition of the growth factors, and the formation of cystic EBs was observed after 7-10 days, similarly to cavitated EBS formed from hESCs following 10 days in culture [Itskovitz et al, 2000], and cystic EBs after 14-20 days. Within the EBs formed from hESCs or iPS cells there were cell types representative of the three embryonic germ layers typical of iPS cells differentiation (data not shown).

(300) iPS cells or hESCs which are cultured in dynamic suspension maintain their pluripotency in vivo—Pluripotency of iPS cells or hESCs cultured in dynamic suspension demonstrated in vivo by teratoma formation. Cells were cultured in static suspension for at least 20 passages and then in dynamic suspension for additional 10 passages and then were injected into SCID Beige mice. Following 10 weeks of injection into the mice tumors were formed. Within these teratomas, tissues representative of the three germ layers were observed (data not shown).

(301) This study presents a novel approach for culturing undifferentiated iPS cells or human ESCs using either defined 2D culture system or suspension cultures. The present inventors demonstrate that under these conditions two iPS cell lines, one derived from adult fibroblasts and one derived from foreskin fibroblast could be grown and expanded through many passages while maintaining their features including pluripotency and stable karyotypes. When iPS cells are transferred to suspension in the presence of a differentiating medium (e.g., DMEM/F12 supplemented with 10% fetal bovine serum (FBS), 10% KNOCKOUT™ serum replacement, 2 mM L-glutamine, 0.1 mM β -mercaptoethanol, and 1% non-essential amino acid stock), they spontaneously form embryoid bodies (EBs). On the other hand, using the tested culture systems (e.g., in the presence of the CM100F, CM100Fp, yF100 or yFL3 culture media) iPS cells spontaneously form spheroids consisting undifferentiated cells.

(302) This is the first description of a method for continuous expansion of undifferentiated iPS in 3D suspension and shaking cultures, which could be adequately applied for large-scale cell production.

(303) The inventors present for the first time a suspension culture system for expansion of undifferentiated iPS, based on serum free medium and defined growth factors. This suspension culture system utilizes either Petri dishes, shaking Erlenmeyer, or spinner flasks. Two iPS cell lines from adult skin and newborn foreskin fibroblast were cultured according to the novel method of the invention as small spheroids which maintain all typical ESC/iPS cells features following prolonged culture of over 25 passages (86 doublings), including stable karyotype and pluripotency. These results demonstrate that culturing iPS cells in a defined medium without feeder layer using 3D culture is possible.

(304) In addition, when applied onto a dynamic system for one month, the number of cell clumps of both hESCs and human iPS cells increased in folds while maintaining the cells unique characteristics. These results render the proposed suspension system suitable for both the routine culture of iPS cells or hESCs in 3D and for mass production of iPS cells and hESCs for therapeutic ends.

(305) The teachings of the invention present scalable, reproducible and controlled culture systems. These results present a significant progress towards the desired end goal of obtaining a facilitator method for large-scale culture of undifferentiated iPS cells and hESCs needed for both clinical and industrial uses.

(306) Although the invention has been described in conjunction with specific embodiments thereof, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and broad scope of the appended claims.

(307) It is the intent of the applicant(s) that all publications, patents and patent applications referred to in this specification are to be incorporated in their entirety by reference into the specification, as if each individual publication, patent or patent application was specifically and individually noted when referenced that it is to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention. To the extent that section headings are used, they should not be construed as necessarily limiting. In addition, any priority document(s) of this application is/are hereby incorporated herein by reference in its/their entirety.

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Claims

1. A culture medium comprising a cytokine consisting of an IL6RIL6 chimera at a concentration range of about 50-200 picogram per milliliter (pg/ml), wherein said IL6RIL6 chimera includes the amino acid sequence set forth by SEQ ID NO: 39, wherein the culture medium is serum-free and capable of maintaining pluripotent stem cells in an undifferentiated state in the absence of feeder cell support for at least 5 passages.
2. The culture medium of claim 1, wherein said culture medium further comprises serum replacement.
3. The culture medium of claim 1, wherein said concentration of said IL6RIL6 chimera is between about 90 pg/ml to about 120 pg/ml.
4. The culture medium of claim 1, wherein said concentration of said IL6RIL6 chimera is about 100 pg/ml.
5. The culture medium of claim 1, wherein said culture medium further comprises basic fibroblast growth factor (bFGF).
6. A cell culture comprising pluripotent stem cells and the culture medium of claim 1.
7. The cell culture of claim 6, wherein the cell culture is feeder cells free.
8. The cell culture of claim 6, wherein said pluripotent stem cells are embryonic stem cells.
9. The cell culture of claim 6, wherein said pluripotent stem cells are induced pluripotent stem (iPS) cells.
10. The cell culture of claim 6, wherein said pluripotent stem cells are human pluripotent stem cells.
11. The cell culture of claim 6, being a suspension culture.
12. The culture medium of claim 1, wherein the culture medium is capable of maintaining a population of cells consisting of induced pluripotent stem cells having a normal karyotype for at least 15 passages.
13. A method of deriving an embryonic stem cell line, comprising (a) obtaining an embryonic stem cell from a pre-implantation stage blastocyst, post-implantation stage blastocyst and/or a genital tissue of a fetus; and (b) culturing said embryonic stem cell in the culture medium of claim 1; thereby deriving the embryonic stem cell line.
14. A method of deriving an induced pluripotent stem cell line, comprising (a) inducing a somatic cell to a pluripotent stem cell; and (b) culturing said pluripotent stem cell in the culture medium of claim 1; thereby deriving the induced pluripotent stem cell line.
15. A method of generating lineage-specific cells from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells in the culture medium of claim 1, to thereby obtain expanded, undifferentiated stem cells; (b) subjecting said expanded, undifferentiated stem cells to culturing conditions suitable for differentiating and/or expanding lineage specific cells; thereby generating the lineage-specific cells from the pluripotent stem cells.
16. A method of generating embryoid bodies from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells in the culture medium of claim 1, to thereby obtain expanded, undifferentiated pluripotent stem cells; and (b) subjecting said expanded, undifferentiated pluripotent stem cells to culturing conditions suitable for differentiating said stem

cells to embryoid bodies; thereby generating the embryoid bodies from the pluripotent stem cells.

17. A method of generating lineage-specific cells from pluripotent stem cells, the method comprising: (a) culturing the pluripotent stem cells in the culture medium of claim 1, to thereby obtain expanded, undifferentiated pluripotent stem cells; (b) subjecting said expanded, undifferentiated pluripotent stem cells to culturing conditions suitable for differentiating said expanded, undifferentiated stem cells to embryoid bodies; and (c) subjecting cells of said embryoid bodies to culturing conditions suitable for differentiating and/or expanding lineage specific cells; thereby generating the lineage-specific cells from the pluripotent stem cells.

18. A method of expanding and maintaining pluripotent stem cells in an undifferentiated state, the method comprising culturing the pluripotent stem cells in the culture medium of claim 1, thereby expanding and maintaining the pluripotent stem cells in the undifferentiated state.

19. The method of claim 18, wherein said culturing is effected on a matrix.

20. The method of claim 18, wherein said culturing is effected in a suspension culture.

21. The method of claim 18, wherein said culture medium is protein carrier-free.
